

Master Thesis

Developing a Framework for Computing Cartograms

Emil Künstner
I6377110

Thesis submitted in partial fulfillment
of the requirements for the degree of
Master of Science of Artificial Intelligence
at the Department of Advanced Computing Sciences
of the Maastricht University

Thesis Committee:

Dr. M.F.M. Sondag
Dr. M. Mihalak

Maastricht University
Faculty of Science and Engineering
Department of Advanced Computing Sciences

August 25, 2026

Abstract

Cartograms are used a lot to visualize geospatial statistics, but existing cartogram styles, such as Demers, Dorling, non-contiguous, and contiguous cartograms, are typically produced by isolated, style-specific algorithms, making it difficult to compare types or reason about their trade-offs within a common framework. This thesis asks whether a single mathematical optimization problem can unify cartogram generation across these four types. A convex quadratic program is formulated in which each cartogram type arises as a distinct parameter regime of the same underlying model, built around a homothetic vertex transform that keeps region-area control convex and a set of soft topological constraints that approximate, without guaranteeing, polygon non-intersection.

The resulting framework is benchmarked against established specialized implementations on six real-world datasets spanning electoral, economic, and demographic statistics across the United States, India, China/Taiwan, the Netherlands, Germany, and the world. Evaluated on statistical, geographical, and topological quality criteria drawn from the cartogram literature. Demers, Dorling, and non-contiguous cartograms produced by the framework match or exceed the quality of their specialized counterparts. Contiguous cartograms, on the other hand, underperform compared to both baselines, a direct consequence of polygon non-intersection being non-convex and therefore only softly enforceable. Across every mode, the unified framework is substantial slower than the specialized methods it is compared against.

These findings indicate that unification, though not complete, of cartogram generation within a single convex framework is achievable, with runtime and contiguous-mode topology remaining the framework’s principal limitations.

Statement on the Use of AI

AI tools were used in an advisory capacity during this thesis, for wording suggestions and minor coding assistance, with all output reviewed and verified by the author, who takes full responsibility for the thesis's content and conclusions.

Contents

1	Introduction	5
2	Related Work	7
2.1	Cartogram Types and Algorithms	7
2.2	Creation Algorithms	8
2.3	The Challenge of Contiguous Cartograms in Optimization	8
2.4	Quality Criteria for Cartogram Evaluation	9
3	Methodology	11
3.1	Map Quality Criteria	11
3.2	Notation	12
3.3	Parameters	13
3.4	Variables	15
3.5	Preprocessing	15
3.6	Area-Constrained Polygon Scaling	16
3.7	Shape-Constraints	17
3.7.1	Centre Placement	17
3.7.2	Vertex Movement	17
3.7.3	Shape-Deformation	17
3.8	Topology-Constrained Intersecting Regions	17
3.9	Definitions	18
3.10	Constraints	18
3.11	Optimisation Problem	19
3.11.1	Energy terms	19
3.12	Post-processing	20
3.12.1	Raster discretisation	20
3.12.2	Disputed-pixel resolution and gap filling	21
3.12.3	Raster area equalisation	21
3.12.4	Circle declumping (Dorling only)	21
3.13	Recommender System	21
3.13.1	Sliders and base preset	22
4	Experiments	23
4.1	Reference Implementations	23
4.2	Datasets	23
4.3	Comparative Protocol	24
4.4	Evaluation Metrics	24

4.4.1	Statistical Accuracy: Cartographic Error	24
4.4.2	Geographical Accuracy: Shape Preservation	25
4.4.3	Geographical Accuracy: Positional Fidelity	25
4.4.4	Topological Accuracy: Adjacency Error	25
4.4.5	Topological Integrity: Overlap and Self-Intersection	26
4.4.6	Topological Integrity: Gap Error	26
4.4.7	Complexity and Runtime	26
4.5	Results	27
4.5.1	Computing Time	27
4.5.2	Per-Dataset Comparison	27
4.5.3	Postprocessing	36
5	Discussion	46
6	Conclusion	49
6.1	Limitations	49
6.2	Future Work	50
7	Appendix	56
7.1	Quality Criteria	56

Chapter 1

Introduction

Many datasets contain data related to geospatial information, like the population of a region. Visualizing such information in an effective way poses a challenge [1]. One option for visualizing geospatial data is the use of choropleth maps. Those visualize statistical information over the color hue, saturation or luminance. Choropleth maps have limitations. It is difficult to compare colour hue, saturation and luminance, and regions with a small geographical area remain small no matter the statistical value. This makes them hard to spot, even though they may be important. [1].

An alternative way to visualize statistical values with geospatial information are Cartograms. Instead of using colors to represent statistical information, regions get scaled according to the corresponding statistical value, [2] addressing one of the mentioned problems of choropleths making regions with high statistical importance visually more prominent. Holbrook and Cartwright found area cartograms to be more effective for simple messages [3]. Other studies have reported different results depending on the specific cartogram and comparison method used [4], suggesting that the effectiveness of a cartogram depends not only on the data but also on the design and task for which it is used.

Changing the areas of regions with complex geographical shapes usually leads to problems fitting the resulting regions back together and with that preserving a readable map. To address this problem several types have been introduced and are in use, including Demers cartograms [5], Dorling cartograms [6] and mosaic cartograms [7, 8]. These types emphasize different cartographic quality criteria, such as shape preservation, adjacency, continuity, or readability, making the choice of the cartogram type important for correct data interpretation [2].

Researchers, analysts, and communicators who use cartograms must consider these trade-offs when selecting an appropriate representation [9]. However, most existing cartograms are produced by isolated algorithms [2]. This makes it difficult to compare different cartogram types within a common framework or to interactively explore the trade-offs between different cartographic properties.

A unified Framework would make the Cartogram Types outcomes of the same algorithm given different parameters. Existing Cartogram algorithms use various methods with force directed/physics based algorithms being very prominent [2]. Mathematical optimization has also been applied to cartogram generation, for example in linear-programming-based approaches for rectangular and Demers cartograms [10]. Such approaches are particularly suitable for a unified framework

because desired cartographic properties can be expressed explicitly through objectives and constraints, allowing different cartogram types to emerge as different optima of the same optimization space.

Instead of another isolated cartogram algorithm, this work presents a cartogram framework which is capable of creating various cartogram styles within one mathematical optimization problem. Various types are optima of the same problem given different parameters. Concretely, properties such as “regions are drawn as circles” (Dorling) or “regions keep their original shared borders” (contiguous) are not implemented as separate pieces of code, but arise from switching individual objective terms and constraints on or off within one convex quadratic program. This also provides an opportunity to explore other Cartograms of the defined optimization space.

These considerations lead to the following research questions, which structure the remainder of this thesis:

1. Can we use mathematical programming or force-directed layout to implement a fast cartogram generation framework for multiple types of cartograms?
2. Which cartographic quality criteria are necessary and sufficient to represent major cartogram types within a unified framework, and what dependencies or trade-offs exist between these criteria?
3. How can the identified cartographic quality criteria be formally encoded as linear constraints or force-based objective functions within a unified optimization model?
4. How does the proposed unified framework compare to specialized cartogram algorithms in terms of computational efficiency and quality metrics across different datasets?
5. Can an automated recommendation mechanism be designed to help select a suitable cartogram type based on dataset characteristics and cartographic quality criteria?

In answering these questions, this thesis makes the following contributions. First, it formulates a single convex quadratic program that reproduces four established cartogram types: Demers, Dorling, non-contiguous and contiguous cartogram, as different optima of the same optimization problem. Secondly, it maps established cartographic quality criteria onto the cartogram types that satisfy them, to motivate which criteria the framework needs to encode as objective terms or constraints described in (Chapter 3). Third, it benchmarks the resulting framework against established reference implementations, captured in two packages the R cartogram package [11] and cartogram-cpp [12], on real-world datasets covering the USA, India and China/Taiwan, using a shared set of statistical, geographical and topological quality criteria [9, 13] described in (Chapter 4). Finally, it identifies and documents the main obstacle to a fully unified framework, that being polygon non-intersection cannot be expressed as a convex constraint, and discusses how this limits contiguous-cartogram quality and what directions could address it.

Chapter 2 reviews existing cartogram types, creation algorithms, and evaluation criteria, and identifies the research gap this thesis addresses. Chapter 3 describes the designed optimization framework. Chapter 4 describes the experimental setup and evaluation metrics used to benchmark it. The thesis closes with a discussion of the results and their implications, followed by conclusions, limitations and future work.

Chapter 2

Related Work

Chapter 1 argued that a single optimization-based framework could unify cartogram generation, which is currently spread across many isolated, style-specific algorithms. This chapter builds the case for that argument in detail. Sections 2.1 and 2.2 survey which cartogram styles and creation algorithms exist and why their differences matter for a unifying framework, Section 2.3 explains why, of the available algorithmic approaches, contiguous cartograms in particular have resisted a convex formulation, and Section 2.4 reviews the quality criteria used to evaluate cartograms, which this thesis later reuses for its own experiments in Chapter 4.

Cartogram like visualization have been used since the 1800s [2]. In 1900s most of the common ones where formally defined including Raisz’s rectangular statistical cartogram [14] and Dorling’s area cartogram [6]. The development of automatically generated cartograms, including continuous deformation approaches in the 1970s [15], further contributed to the development of modern cartogram algorithms.

2.1 Cartogram Types and Algorithms

Nusrat and Kobourov gathered various cartogram implementations [2] and defined the following main styles:

1. **Dorling cartograms** [6] represent the regions as circles scaled proportionally to the data value, sacrificing topological contiguity for shape simplicity.
2. **Demers cartograms** [5, 10, 16] show regions as squares.
3. **Contiguous cartograms** [15, 17, 18, 19, 20, 6, 21, 22, 23, 24, 25, 8, 26] preserve the connectivity of the original map while deforming the region boundaries continuously.
4. **Non-contiguous cartograms** [27] scale regions in place without attention to topology.
5. **Rectangular cartograms** [28, 29, 30, 26, 10, 16] represent regions as rectangles, with dense packing.
6. **Mosaic cartograms** [7, 8] use tile grids to represent regions.

These six styles do not represent every cartogram type defined so far, but they are the set most consistently used and compared against one another in the literature, which is why Nusrat and

Kobourov’s survey [2] treats them as the canonical categorization. This set matters for the present thesis because it defines the target space the proposed framework is measured against: rather than inventing a new cartogram style, this work asks how many of the defined styles above can be expressed as different parameter settings of one optimization problem. Of the six styles, this thesis implements and evaluates four: Demers, Dorling, non-contiguous and contiguous cartograms, because together they span the two design axes identified as most consequential for the quality-criteria trade-offs discussed in Section 2.4, how much a region’s original shape is simplified (kept as-is, or replaced by a circle or square) and whether topological contiguity between neighbours is preserved or sacrificed. Mosaic and rectangular cartograms were left out because they replace the continuous, per-vertex polygon representation used by the other four styles with a fundamentally discrete tiling or packing problem, which does not fit directly into the same continuous, per-vertex optimization variables used elsewhere in this framework (see Chapter 3) extending the framework to cover them is discussed as future work.

2.2 Creation Algorithms

The algorithms for generating these cartograms can be grouped into three broad categories:

Mathematical programming have been applied to rectangular cartograms [31, 32] and Demers cartograms [10, 16]. These methods formulate desired cartographic properties as objectives and constraints that are solved through an optimization procedure.

Force-directed and physics-based approaches form the largest class. Dorling’s original circle-based method employs a force-directed approach [6]. Contiguous cartograms have been created using rubber-sheet algorithms [33, 24, 25] as well as other physics-based approaches [15, 19].

Other approaches include the mosaic-based method of Cano et al. [8] and the diffusion-based algorithms of Gastner and Newman [21]

This categorization is relevant for the framework proposed in this thesis because it outlines the space of possible implementation strategies and explains why one of them was chosen over the others. Force-directed and physics-based methods are intuitive to implement and currently form the largest class, but their iterative, simulation-style updates make it hard to reason formally about which quality criteria are being traded off against which, or to combine several cartogram styles inside one coherent formulation with modifiable weights.

Mathematical-programming approaches, by contrast, state the desired properties explicitly as an objective function and a set of constraints, which is exactly the property this thesis relies on to unify several cartogram types in a single problem (Chapter 3).

However, existing mathematical-programming-based approaches have so far covered only a subset of cartogram types. The next section discusses why contiguous cartograms, in particular, remain difficult to represent within a convex optimization framework.

2.3 The Challenge of Contiguous Cartograms in Optimization

Section 2.2 showed that mathematical-programming approaches are well suited to expressing cartographic quality criteria explicitly as objectives and constraints, which is why this thesis

chose them over a physics-based approach. This section explains, in more detail, why, despite that suitability, no LP-based framework has so far covered contiguous cartograms, the style in which regions are deformed continuously from their original shape in the map. This gap is the central technical obstacle that the framework in Chapter 3 has to work around.

Existing mathematical programming based cartogram methods are restricted to non-contiguous types such as Demers and rectangular cartograms [10, 32]. A shared limitation across LP-based approaches is that polygon non-intersection cannot be encoded as a convex constraint: the set of non-intersecting polygon configurations is non-convex, and enforcing separation between polygons with moving vertices introduces bilinear terms in the optimisation variables that violate the disciplined convex programming (DCP) requirements of standard solvers such as CLARABEL [34].

Several $O(m + n)$ algorithms exist for computing the intersection of two convex polygons on fixed geometry [35, 36, 37, 38]. O’Rourke et al. [35] advance two boundary-chasing pointers using halfplane membership tests. Toussaint [36] decomposes the problem into convex hull construction and sail polygon triangulation, offering a conceptually simpler correctness proof. While these algorithms are efficient for post-hoc intersection detection, they operate on fixed vertex coordinates and cannot be embedded as constraints in an optimisation problem where vertex positions are decision variables. No unified LP framework covering contiguous cartogram types currently exists in the literature, and polygon non-intersection in continuous optimisation with moving vertices remains an open problem.

This open problem directly motivates the modelling choice made in Chapter 3. Rather than attempting to enforce full polygon non-intersection as a hard constraint, the framework instead uses a soft, shared-vertex-locking formulation for contiguous cartograms. This approach does not solve the general problem of polygon non-intersection. Instead, it investigates how far a unified convex formulation can be extended towards contiguous cartograms and evaluates where the resulting compromise succeeds and where it breaks down.

2.4 Quality Criteria for Cartogram Evaluation

A recurring criticism in the cartogram literature is that generation methods are usually justified by visual inspection alone. Alam, Kobourov, and Veeramoni [9] and Nusrat, Alam, and Kobourov [13] address this by proposing a set of quality criteria: statistical accuracy, geographical (shape and position) accuracy, and topological accuracy, rather than a single aggregate score.

To measure performance of Cartograms so called Quality Criteria were defined [9, 39, 2, 29, 40] to evaluate cartograms objectively instead subjective impressions. The following nine have been identified as useful for evaluating cartograms [16, 9], which can be categorized into the three broader categories: statistical accuracy (does region size match the data value), geographical accuracy (shape and position), and topological accuracy (adjacency and readability). Figure 2.1 illustrates each criterion with a small synthetic example.

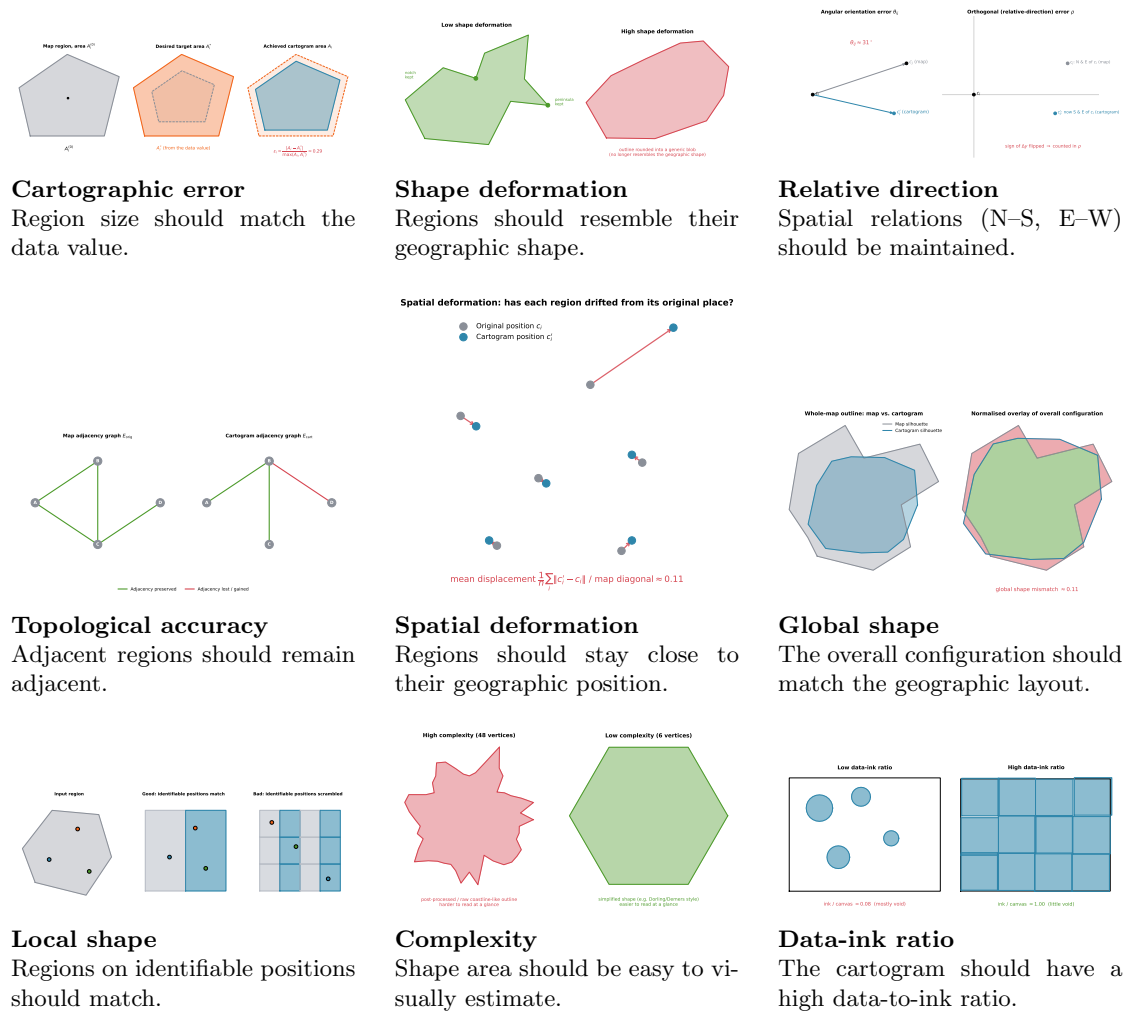


Figure 2.1: The nine cartogram quality criteria used to evaluate the framework (Section 2.4).

Chapter 3

Methodology

This chapter discusses the methodology of a unified Cartogram Framework. Starting with some preliminary work of mapping quality criteria to cartogram styles in Section 3.1, followed by a deep dive into notations 3.2, parameters 3.3 and variables 3.4 used throughout the methodology. Followed by a walk through the Framework in the Sections 3.5, 3.6, 3.7, 3.8 with the formulas of the constraints 3.10 and objectives 3.11 just after. Closing with the post-processing in Section 3.12 and the recommender system in section 3.13

3.1 Map Quality Criteria

To get a good insight on which Quality Criterias are important for which features the Quality Criterias were mapped against the Cartogram types. Every Cartogram type got a score of 0, 0.5 or 1 for how much they would adhere a Quality Criteria where 0 means they perform bad e.g. have the full error, 0.5 means they manage to do something in the right direction and 1 means full implementation of the Quality Criteria except certain very specific edge cases e.g. Topology in regions encapsulated by a fully landlocked state with too little tiles in a Mosaic Cartogram. A different variant of this mapping was done by Markowska [citemarkowska19:cartograms-classification-terminology](#).

This mapping motivates the choice, in Section 3.11, to encode cartographic error, shape deformation, spatial deformation and topological accuracy as explicit objective/constraint terms, since Table 3.1 shows these are exactly the criteria on which the four implemented types differ most from a normal map.

Table 3.1: Qualitative adherence of each cartogram type to the nine quality criteria of Section 2.4, scored 0 (poor), 0.5 (partial), or 1 (full).

Cartogram Type		Cartographic Error	Shape Deformation	Preservation of Rel. Direction	Topological Accuracy	Spatial Deformation	Global Shape	Local Shape	Complexity	Data Ink Ratio
Normal Map		0	1	1	1	1	1	1	0.5	0.5
Mosaic	Cartogram	0.5	0.5	0.5	1	1	0.5	0.5	0.5	0.5
Demers	Cartogram	1	0	1	1	0.5	0.5	0	1	0.5
Dorling	Cartogram	1	0	0.5	0.5	0.5	0.5	0	1	0
Non-contiguous Cartogram		1	1	1	0	1	1	1	0.5	0
Contiguous Cartogram		1	0.5	0.5	1	0.5	0.5	0.5	0.5	0.5
Rectangular Cartogram		1	0	0.5	1	0.5	0	0	1	1
Gridmap		1	0	1	0.5	1	0.5	0	1	0.5

3.2 Notation

Throughout this thesis, map refers to the original input polygons (the geographic boundaries before any transformation), and cartogram refers to the transformed output polygons produced by the optimization problem or other algorithms. The table below uses this map/cartogram terminology instead of generic before/after language.

Table 3.2: Summary of Notation used throughout this Thesis.

Quantity	Symbol	Description
Region set	$\mathcal{R}$	$\{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_i, \dots, \mathcal{R}_N\}$
Input polygon	P_i	$\{v_{i,1}, v_{i,2}, \dots, v_{i,k}, \dots, v_{i,n_i}\}$
Vertex k of P_i	$v_{i,k}$	$(x_{i,k}, y_{i,k})^\top \in \mathbb{R}^2$
Original area	$A_i^{(0)}$	Shoelace area of region i in the map
Target area	A_i^*	Normalised area region i should have in the cartogram (Eq. 3.1)
Target scale	s_i	$\sqrt{A_i^*/A_i^{(0)}}$ (Eq. 3.2)
Reference centre	$c_i \in \mathbb{R}^2$	$= \frac{1}{n_i} \sum_k v_{i,k}$: Fixed point, centroid of region i in the map
Offset vector	$d_{i,k} \in \mathbb{R}^2$	$v_{i,k} - c_i$ (Eq. 3.3)
Cartogram vertex	$v'_{i,k} \in \mathbb{R}^2$	Position of vertex k of region i in the cartogram (Eq. 3.4)
Horizontal set	$H \subseteq \mathcal{R}^2$	Pairs with i left of j
Vertical set	$V \subseteq \mathcal{R}^2$	Pairs with i below j
Width proxy	w_{ij}	$\frac{1}{2}(\sqrt{A_i^*} + \sqrt{A_j^*})$
Gap	g_{ij}	0 if adjacent, ε otherwise
Dir. weight	b_{ij}	$b \cdot a_{ij}$, $a_{ij} = 1$ if $(i, j) \in T$ else 0.1
ℓ^∞ rad.	$M_{i,k}$	$\max(d_{i,k}^x , d_{i,k}^y)$

3.3 Parameters

Table 3.3: Summary of Parameters used in the optimization problem.

Parameter	Symbol	Description
Shape weight	λ_s	Weight on shape-deformation term
Area weight	λ_a	Weight on cartographic-error term
Centre weight	λ_c	Weight on centre-fidelity term
Topology weight	λ_t	Weight on pairwise topology term
Contiguous weight	λ_{con}	Weight on soft shared-vertex penalty
Order weight	λ_{order}	Weight on soft ordering penalty (0 disables ordering)
Mean scale	$\lambda_{\text{mean_scale}}$	Weight on mean scale area term
Shape/target blend	$\gamma \in [0, 1]$	1 = pure deformation, 0 = pure E^{target}
L1/L2 blend	$\beta \in [0, 1]$	1 = pure L1, 0 = pure L2 deformation
Diagonal slope	α	Slope for directional deviation (3.12)
Non-adj. gap	ε	Minimum gap for non-adjacent pairs
Dir. scale	b	Scale factor for b_{ij}
Min. scale	$t_{\min}$	Lower bound on per-vertex scale
Max. scale	$t_{\max}$	Upper bound on per-vertex scale (prevents runaway)
Target centre	$c_i^* \in \mathbb{R}^2$	Prescribed centre position for region i
Shared-vertex set	S_{ij}	Index pairs (k, l) of shared vertices between i, j

3.4 Variables

Table 3.4: Additional variables used in the optimization problem.

Variable	Symbol	Description
Centre displacement	$\Delta_i \in \mathbb{R}^2$	Translation of region i 's centre
Moved centre	$m_i \in \mathbb{R}^2$	$m_i = c_i + \Delta_i$
Per-vertex scale	$t_{i,k} \in \mathbb{R}$	Radial scale of vertex k in region i
Abs. deviation aux.	$z_{i,k} \in \mathbb{R}_{\geq 0}$	$z_{i,k} \geq t_{i,k} - 1 $ (L1 linearisation)
Square half-side	$r_i \in \mathbb{R}_{\geq 0}$	Target ℓ^∞ -radius (Demers mode only)
Horiz. excess	$h_{ij} \in \mathbb{R}_{\geq 0}$	Horizontal excess distance for pair $(i, j) \in T$
Vert. excess	$v_{ij} \in \mathbb{R}_{\geq 0}$	Vertical excess distance for pair $(i, j) \in T$
Dir. deviation	$\delta_{ij} \in \mathbb{R}_{\geq 0}$	Directional (diagonal) deviation for $(i, j) \in T$
Horiz. order slack	$s_{ij}^h \in \mathbb{R}_{\geq 0}$	Ordering violation for pair $(i, j) \in H$
Vert. order slack	$s_{ij}^v \in \mathbb{R}_{\geq 0}$	Ordering violation for pair $(i, j) \in V$

3.5 Preprocessing

Let $\mathcal{R} = \{\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_i, \dots, \mathcal{R}_N\}$ be the set of N input regions. Each region i is given as an ordered polygon $P_i \in \mathbb{R}^{n_i \times 2}$.

Area normalization. Raw target areas $\hat{A}_i$ are rescaled so that their sum equals the total original map area:

$$A_i^* = \hat{A}_i \cdot \frac{\sum_{j=1}^N A_j^{(0)}}{\sum_{j=1}^N \hat{A}_j}, \quad (3.1)$$

Without this step, the target areas supplied by an arbitrary dataset (e.g. GDP figures whose sum bears no relation to the geographic area of the map) would force every region to grow or shrink, rather than only redistributing the fixed total area among regions the way a cartogram is meant to.

Reference centre and target scale. The (fixed) reference centre of region i is its centroid, here calculated by: $c_i = \frac{1}{n_i} \sum_k v_{i,k}$. The uniform target scale factor is:

$$s_i = \sqrt{A_i^* / A_i^{(0)}}. \quad (3.2)$$

Offset vectors. For every vertex k of region i , the constant offset vector $d_{i,k}$:

$$d_{i,k} = v_{i,k} - c_i \in \mathbb{R}^2 \quad (3.3)$$

fixes the direction from c_i to that vertex before optimisation begins. Because $d_{i,k}$ is never re-derived from the moving vertices, each vertex is restricted to move only along the fixed ray through c_i and its own original position, leaving a one-dimensional freedom per vertex rather than a free position in $\mathbb{R}^2$ 3.6.

Adjacency. $T \subseteq \mathcal{R}^2$ denotes the set of geographically neighbouring pairs (sharing at least one vertex within tolerance τ_{adj}). $H, V \subseteq \mathcal{R}^2$ are computed independently of T . H contains every pair for which region i lies left-of region j in the original map, V every pair for which i lies below j , whether or not i and j are adjacent. All three sets are computed once and kept fixed throughout the optimisation.

3.6 Area-Constrained Polygon Scaling

Cartograms express the statistical information through the sizes of their regions $\mathcal{R}$. If such a Region is not simplified to a simple form like it is done in Dorling’s or Demers cartograms, there has to be a way to scale Polygons to have a desired area.

For Dorling’s and Demers the Framework gets a head start. Since the goal are perfect sized circles and squares the original regions are switched with a circle or square respectively.

Assuming a Polygon $P_i = \{v_{i,1}, v_{i,2}, \dots, v_{i,k}, \dots, v_{i,n_i}\}$ with area $A_i^{(0)}$ we aim to compute a new Polygon $P'_i = \{v'_{i,1}, v'_{i,2}, \dots, v'_{i,k}, \dots, v'_{i,n_i}\}$ such that the area of $P'_i = A_i^*$ while minimizing shape distortion.

Directly optimizing vertex coordinates for a specific area presents two primary challenges:

1. **Non-convexity of the Area Function:** The Shoelace formula [41, 42] defines the area of a polygon as a bilinear form of its vertex coordinates. Since the area calculation involves products of variables (e.g., $x_{i,k}y_{i,k+1} - x_{i,k+1}y_{i,k}$), the resulting equality constraint is non-convex, making it incompatible with standard convex optimization solvers [34].
2. **Shape Preservation:** Allowing free movement of vertices in $\mathbb{R}^2$ leads to non-convexity, so the vertex movement of P_i must be restricted.

To address these issues, rather than treating (x, y) coordinates as independent variables, we applied a homothetic transform [43, 44] by defining a fixed reference point, the centroid $\mathbf{c}_i = \frac{1}{n_i} \sum_k \mathbf{v}_{i,k}$, and restrict vertex movement along the vectors between $\mathbf{c}_i$ and the vertices $\mathbf{v}_{i,k}$ of P_i . We introduce a vector $d_{i,k}$ already introduced in the previous section 3.3 of scaling factors $\mathbf{t}_{i,\mathbf{k}} \in \mathbb{R}^n$, where each $t_{i,k}$ represents the radial displacement of vertex $\mathbf{v}_{i,k}$.

For the non-convexity of the shoelace formula we employ a radial scaling surrogate. Since a uniform scaling factor s scales area by s^2 , the target mean scale s_i ((3.2)) is defined so that a region whose vertices scale uniformly by s_i would exactly hit its target area A_i^* . Rather than constraining the true, non-convex area directly, the mean of the vertex-wise scales $t_{i,k}$ is getting optimized to equal s_i , and the resulting deviation is penalised in the objective as the per-region area energy E_i^{area} ((3.13)).

Two further constraints keep this surrogate well-behaved. First, because the mean-scale equality only constrains the average of a region’s vertex scales, nothing on its own prevents a small number of vertices from growing arbitrarily large while others shrink to compensate and the mean still holds. This was experienced during the development phase so that the scale bounds $t_{\min} \leq t_{i,k} \leq t_{\max}$ ((3.6)) were introduced to cap this per-vertex spread. Second, $t_{\max}$ in particular acts as a safety valve for datasets with extreme area ratios (Section 3.13’s robustness slider adjusts exactly this bound). Without an upper bound, a region whose target area is disproportionately large relative to its original size could lead the solver toward huge per-vertex scale spikes.

3.7 Shape-Constraints

After having a way to resize polygons to a target area, the next step towards Cartograms is the placing of the regions. Therefore the dependence of every vertex from the center is used, since through that the movement of the center moves the whole polygon.

3.7.1 Centre Placement

Every region is free to move as a whole via a centre-displacement variable Δ_i , moving its centroid c_i toward a prescribed target centre c_i^* . This translation is penalised in the objective by the centre-fidelity energy E_i^{centre} (3.14), weighted by λ_c . Especially for Dorling and Demers modes, c_i^* is used. Here c_i is the region's original geographic centroid, so λ_c directly trades positional fidelity against the freedom the solver needs to resolve overlap and topology elsewhere in the problem (Table 3.5, position_accuracy).

3.7.2 Vertex Movement

Combining the centre translation Δ_i (Section 3.7.1) and the per-vertex radial scale $t_{i,k}$ (Section 3.6) with the fixed offset $d_{i,k}$ gives the cartogram vertex position (3.4). Because $d_{i,k}$ is constant, (3.4) is affine in $(\Delta_i, t_{i,k})$. The property every later constraint and objective term relies on to remain DCP-compliant.

3.7.3 Shape-Deformation

The per-vertex scale $t_{i,k}$ also determines how much a region's shape is distorted, independent of its area. A region whose vertices all share the same scale factor is just enlarged or shrunk, while a region whose vertices scale unevenly is modified in its shape. The shape-energy term E_i^{shape} ((3.16)) penalises this, blended between an L1 and an L2 measure of deviation from $t_{i,k} = 1$. The L1 branch requires an extra non-negative variable $z_{i,k}$ (3.7) linearising $|t_{i,k} - 1|$ into two linear inequalities.

3.8 Topology-Constrained Intersecting Regions

Sections 3.6–3.7 scale, move and deform each region independently. Without further constraints, nothing stops two regions from overlapping or from drifting so far apart that their original adjacency is lost. Exactly the two failure modes Section 2.3 identified as unavoidable in a fully convex formulation. This section describes the pairwise constraints that aim to keep neighbouring regions neighbours and to prevent overlapping even though true non-intersection cannot be guaranteed.

Horizontal ordering (H -pairs) and vertical ordering (V -pairs). For every pair of regions where one lies strictly left-of or below the other in the original map, the corresponding moved centres are constrained to preserve that ordering, offset by a width proxy w_{ij} ((3.8), (3.9), (3.5)) inspired by Nickel et al. [16]. w_{ij} is hereby sized from both regions' target areas so that two regions large enough to occupy their target areas side by side are not forced into artificial overlap by the ordering constraint alone. Non-adjacent pairs receive an additional margin $g_{ij} = \varepsilon$ ((3.5)) also inspired by Nickel et al. [16].

Excess-distance linearisation (adjacent pairs T). Ordering alone only prevents regions from crossing each other. It says nothing about how far apart two originally-adjacent regions are allowed to drift once that ordering is respected. Two auxiliary non-negative variables, h_{ij} and v_{ij} ((3.10), (3.11)), measure this excess distance along each axis, each implemented as two linear inequalities linearising an absolute value in the same way as $z_{i,k}$ (Section 3.7). These enter the objective directly ((3.20), pairwise topology term), penalising adjacent pairs that separate further than their combined width proxy w_{ij} .

Directional deviation (adjacent pairs T). Excess distance alone can still let a region drift diagonally past its neighbour without being penalised, since h_{ij} and v_{ij} are measured independently per axis. The directional-deviation variable δ_{ij} ((3.12)) instead penalises deviation from the original relative direction between adjacent centres, using the diagonal slope α between c_i and c_j in the original map. The weight $b_{ij} = b \cdot a_{ij}$ is higher for genuinely adjacent pairs ($a_{ij} = 1$) than for the remaining, non-adjacent pairs ($a_{ij} = 0.1$), so directional drift is penalised everywhere but more where regions were originally neighbours. [10, 16]

Contiguous vertex locking. Every vertex shared between two adjacent regions is coupled through the penalty (3.15), added to the objective (3.20) with weight λ_{con} . Because (3.4) is affine, this term is a convex quadratic despite coupling two regions' independent transforms. It is applied to every adjacent shared-vertex pair without restriction.

3.9 Definitions

$$v'_{i,k} = m_i + t_{i,k} d_{i,k} = (c_i + \Delta_i) + t_{i,k} (v_{i,k} - c_i). \quad (3.4)$$

$$w_{ij} = \frac{1}{2} \left(\sqrt{A_i^*} + \sqrt{A_j^*} \right), \quad g_{ij} = \begin{cases} 0 & (i, j) \in T, \\ \varepsilon & \text{otherwise.} \end{cases} \quad (3.5)$$

3.10 Constraints

$$t_{\min} \leq t_{i,k} \leq t_{\max}, \quad \forall i, k, \quad (\text{scale bounds}) \quad (3.6)$$

$$z_{i,k} \geq t_{i,k} - 1, \quad z_{i,k} \geq -(t_{i,k} - 1), \quad \forall i, k, \quad (\text{L1 linearisation}) \quad (3.7)$$

$$s_{ij}^h \geq (w_{ij} + g_{ij}) - (x_j - x_i), \quad \forall (i, j) \in H, \quad (3.8)$$

$$s_{ij}^v \geq (w_{ij} + g_{ij}) - (y_j - y_i), \quad \forall (i, j) \in V, \quad (3.9)$$

$$h_{ij} \geq |x_i - x_j| - w_{ij} + g_{ij}, \quad \forall (i, j) \in T, \quad (3.10)$$

$$v_{ij} \geq |y_i - y_j| - w_{ij} + g_{ij}, \quad \forall (i, j) \in T, \quad (3.11)$$

$$\delta_{ij} \geq |y_i + \alpha(x_j - x_i) - y_j|, \quad \forall (i, j) \in T, \quad (3.12)$$

3.11 Optimisation Problem

3.11.1 Energy terms

$$E_i^{\text{area}} = (1 + \lambda_{\text{mean_scale}}) \left(\frac{1}{n_i} \sum_{k=1}^{n_i} t_{i,k} - s_i \right)^2. \quad (3.13)$$

$$E_i^{\text{centre}} = \|m_i - c_i^*\|^2, \quad (3.14)$$

$$E^{\text{contig}} = \sum_{\substack{(i,j) \in T \\ (k,l) \in S_{ij}}} \frac{\|v'_{i,k} - v'_{j,l}\|^2}{\max(\|c_i - c_j\|^2, 10^{-6})} \quad (3.15)$$

Per-region shape energy

$$E_i^{\text{shape}} = \gamma \left(\beta \sum_{k=1}^{n_i} z_{i,k} + (1 - \beta) \sum_{k=1}^{n_i} (t_{i,k} - 1)^2 \right) + (1 - \gamma) E_i^{\text{target}}, \quad (3.16)$$

where $\gamma \in [0, 1]$ blends deformation against target-shape energy, and $\beta \in [0, 1]$ blends L1 against L2 deformation. The target-shape energy E_i^{target} depends on the cartogram mode:

$$E_i^{\text{target, circle}} = \sum_k \left(t_{i,k} \|d_{i,k}\| - \bar{r}_i \right)^2, \quad \bar{r}_i = \frac{1}{n_i} \sum_k t_{i,k} \|d_{i,k}\|, \quad (3.17)$$

$$E_i^{\text{target, square}} = \sum_k \left(t_{i,k} M_{i,k} - r_i \right)^2, \quad M_{i,k} = \max(|d_{i,k}^x|, |d_{i,k}^y|), \quad (3.18)$$

$$E_i^{\text{target, original}} = 0. \quad (3.19)$$

Objective

$$\begin{aligned}
& \min_{\Delta, t, z, r, h, v, \delta} \quad \left(\begin{array}{l} \text{Per-region shape energy} \\ \lambda_s \sum_{i=1}^n E_i^{\text{shape}} \end{array} \right) + \left(\begin{array}{l} \text{Per-region area error} \\ \lambda_a \sum_{i=1}^n E_i^{\text{area}} \end{array} \right) \\
& + \left(\begin{array}{l} \text{Per-region centre fidelity} \\ \lambda_c \sum_{i=1}^n E_i^{\text{centre}} \end{array} \right) \\
& + \left(\begin{array}{l} \text{Pairwise topology} \\ \lambda_t \sum_{(i,j) \in T} (h_{ij} + v_{ij} + b_{ij} \delta_{ij}) \end{array} \right) \\
& + \left(\begin{array}{l} \text{Contiguous shared-vertex coupling} \\ \lambda_{\text{con}} \sum_{\substack{(i,j) \in T, \\ (k,l) \in S_{ij}}} \frac{\|v'_{i,k} - v'_{j,l}\|^2}{\max(\|c_i - c_j\|^2, 10^{-6})} \end{array} \right) \\
& + \left(\begin{array}{l} \text{Horizontal/vertical ordering} \\ \lambda_{\text{order}} \sum_{(i,j) \in H} (s_{ij}^h)^2 + \lambda_{\text{order}} \sum_{(i,j) \in V} (s_{ij}^v)^2 \end{array} \right)
\end{aligned} \tag{3.20}$$

3.12 Post-processing

Sections 3.6–3.11 formulated area scaling, shape preservation and topological accuracy as convex terms of a mathematical optimization problem, but Section 2.3 showed that true polygon non-intersection is not a convex term. The soft shared-vertex coupling (3.15) and the pairwise ordering terms (3.8)–(3.11) aiming for regions to not overlap without being able to guarantee it. This section describes the raster-based post-processing pipeline that resolves the residual overlaps and gaps geometrically, after the optimization was run. The pipeline runs, in order: disputed-pixel resolution, gap filling, raster area equalisation. For Dorling cartograms a circle-declumping pass is used instead.

3.12.1 Raster discretisation

The remaining passes work on a shared pixel grid rather than on vertex coordinates directly. Let $\Omega \subset \mathbb{R}^2$ be a regular grid of ρ pixels along the longer side of the combined bounding box of all N regions, each pixel of area a_{px} . A label function $L : \Omega \rightarrow \{1, \dots, N, \emptyset\}$ assigns each pixel to the region whose polygon contains it, or $\emptyset$ if no polygon (disputed-overlap pixels are resolved to a single owner by raster priority described in the following section 3.12.2). The measured area of region i on the raster is $\hat{A}_i = a_{\text{px}} |L^{-1}(i)|$, and the relative area error at any point in the pipeline is $\hat{\epsilon}_i = (A_i^* - \hat{A}_i)/A_i^*$.

3.12.2 Disputed-pixel resolution and gap filling

For every pixel $x \in \Omega$ contested by more than one region, ownership is decided by an area-aware nearest-region rule rather than plain proximity. Each contesting region i 's distance to x is shifted by a bias proportional to how far $\hat{A}_i$ currently sits below or above A_i^* ,

$$d'_i(x) = d_i(x) - \psi \cdot \hat{\epsilon}_i \cdot d_{\max}, \quad x \text{ assigned to } \arg \min_i d'_i(x), \quad (3.21)$$

where $d_i(x)$ is the Euclidean distance transform of region i 's undisputed core area, $d_{\max}$ a grid-scale normaliser, and $\psi \in [0, 1]$ the bias strength ($\psi = 0$ recovers plain nearest-region assignment). Under-target regions therefore become more competitive for contested pixels without ever winning a pixel that is not genuinely close to them. Enclosed gaps are resolved by the same rule (3.21) applied to the boundary of the hole. Disputed pixels are always resolved before gaps, since gap-filling assumes a single owner per boundary pixel already exists.

3.12.3 Raster area equalisation

Despite the bias term in (3.21) pixels can be assigned to the region having a higher area overhead. A final pass transfers boundary pixels directly between the most-surplus and most-deficit adjacent regions, recomputing the live deficit after each transferred layer:

$$\text{budget}_{ij} = \min(|\hat{\epsilon}_i| A_i^*, |\hat{\epsilon}_j| A_j^*) / a_{\text{px}}, \quad (3.22)$$

repeated for up to a set times of runs or until every region is within a tolerance of A_i^* . Because pixels only ever move between regions that already share a boundary, this pass cannot reintroduce the overlaps or gaps that were just resolved.

3.12.4 Circle declumping (Dorling only)

For Dorling cartograms, regions are only ever compared as (m_i, r_i) , so the raster pipeline above is unnecessary and overlap is instead removed by the classical Dorling (1996) [6] relaxation: for every still-overlapping pair, both centres are pushed apart along the line joining them by a fraction of the overlap amount,

$$m_i \leftarrow \frac{1}{2}(r_i + r_j - \|m_i - m_j\|) \hat{u}_{ij}, \quad m_j \leftarrow \frac{1}{2}(r_i + r_j + \|m_i - m_j\|) \hat{u}_{ij}, \quad (3.23)$$

with $\hat{u}_{ij}$ the unit vector from m_i to m_j , iterated until no pair overlaps by more than a tolerance or a maximum iteration count is reached. Radii and individual vertices are not changed by this method.

3.13 Recommender System

Sections 2.3–3.11 showed that each cartogram style implemented by the framework corresponds to a particular setting of the parameters. In principle a user could tune these directly, but the amount of parameters makes it difficult to keep the overview and additionally the parameters do not map cleanly onto the quality criteria a user actually cares about (Section 3.1): e.g. increasing λ_c trades off positional fidelity against solver freedom, not something a domain user is likely to reason about. This section addresses RQ5 by building a recommender system on top of the framework. Six named parameters, in the following called sliders to make them distinguishable from the native framework parameter, each corresponding to one interpretable cartographic quality criterion, that map onto concrete framework parameter changes.

3.13.1 Sliders and base preset

For each of the four implemented cartogram types contiguous, Demers, Dorling and non-contiguous the manually tuned parameter setting used to produce the results in Chapter 4.5 are used. All six sliders default to $\kappa = 0.5$, which reproduces the selected cartogram type exactly, so the recommender is a strict generalisation of the fixed preset rather than a replacement for it.

Each slider $\kappa \in [0, 1]$ is associated with one or more original parameters p , each given a low and high target value (ℓ_p, h_p) chosen so that $\kappa = 0$ and $\kappa = 1$ correspond to the most conservative and most aggressive settings of that parameter within its feasible range (Table 3.5). The mapping from a slider position to a parameter value is piecewise linear and passes exactly through the base preset at $\kappa = 0.5$:

$$v_p(\kappa) = \begin{cases} \ell_p + \frac{\kappa}{0.5} (b_p - \ell_p), & \kappa \leq 0.5, \\ b_p + \frac{\kappa - 0.5}{0.5} (h_p - b_p), & \kappa > 0.5, \end{cases} \quad \Delta_p(\kappa) = v_p(\kappa) - b_p, \quad (3.24)$$

where b_p is the base-preset (0.5) value of p . When several sliders act on the same parameter, their deltas are summed before being applied once:

$$p_\tau^{\text{final}} = \text{clip} \left(b_p + \sum_{\text{sliders } \kappa \text{ touching } p} \Delta_p(\kappa), p_{\min}, p_{\max} \right), \quad (3.25)$$

with $[p_{\min}, p_{\max}]$ a hand-specified safety bound per parameter (e.g. $\lambda_a \in [0.1, 5.0]$), independent of the slider's own (ℓ_p, h_p) range, to guarantee the solver is never handed a value known to cause numerical issues. If a parameter named by a slider does not exist for the requested cartogram type: e.g. `lambda_contiguous` only applies to the contiguous mode, the slider's effect on that parameter is skipped.

Slider	Parameter(s) moved	Effect at $\kappa = 1$
size_accuracy	$\lambda_a, \lambda_{\text{mean_scale}},$	lower cartographic error
shape_preservation	λ_s	output stays closer to original form
position_accuracy	λ_c	regions pulled toward original centroid
neighborhoods_kept	λ_t	original neighbours strongly enforced
seamlessness	λ_{con} (contiguous only)	shared borders coincide more tightly
robustness	$t_{\max}$	solver given more slack on extreme area ratios, at the cost of tighter vertex-movement control

Table 3.5: Mapping from recommender sliders to framework parameters.

Chapter 4

Experiments

The unified optimization framework introduced in Chapter 3 covers a whole optimization space of cartograms within a single mathematical formulations. Four of the previously in Section 2.1 introduced cartogram types Demers, Dorling, non-contiguous and contiguous can be reproduced. To assess whether this generality comes at the cost of quality, this chapter benchmarks the framework against four established reference implementations and reports results on a list of quality criteria from other cartogram evaluation literature 2.4. This chapter describes the reference implementations in Section 4.1, the datasets in Section 4.2, and the evaluation protocol in Section 4.3. Section 4.4 defines each metric formally.

4.1 Reference Implementations

Four baselines from two sources were selected to cover the three of the four cartogram types produced by the framework, following the categorization established by Nusrat and Kobourov [2]:

- **The cartogram R package** [11] implements the rubber-sheet distortion algorithm of Dougenik, Chrisman and Niemeyer [33] for contiguous cartograms, as well as Dorling’s circle-based method [6] and Olson’s non-contiguous scaling [27]. Because three cartogram types implemented by the framework (Dorling, non-contiguous and contiguous) have a direct counterpart in this package, it serves as the primary baseline for those modes.
- **Cartogram-cpp** [12] is a C++ re-implementation and extension of the fast flow-based diffusion method of Gastner, Seguy and More [22], itself a performance-oriented successor to the original density-equalizing diffusion algorithm of Gastner and Newman [21]. It is used as second baseline for the contiguous mode, where it represents the current state of the art for flow-based, non-overlapping cartogram generation.

No baseline exists for Demers cartograms in either package and no other implementation following a paper it was found.

4.2 Datasets

Following the benchmark suite used by Gastner, Seguy and More [22] for their flow-based algorithm, six real-world datasets were used: the United States, India, a combined Taiwan/China

map, the Netherlands, Germany and a world map. Geographic boundaries for the USA and for Taiwan/China were taken from the sample data bundled with `cartogram-cpp` [12], boundaries for India were sourced from GADM [45] and for Germany from SimpleMaps [46]. Each dataset pairs its boundaries with a statistical attribute used as the cartogram’s target value: for the USA, the number of electoral college votes per state [47], for Taiwan/China, nominal GDP per province [48], for India, state-wise GDP (SDP) for 2022–23 [49], for the Netherlands [50] and the world [51] population from 2020 is used per state and nations respectively and for Germany population data is used as well but from the year 2024 [52]. USA consists out of 49 regions, India out of 33 regions, Taiwan & China out of 34 regions, the Netherlands out of 12, Germany out of 16 and the world out of 257.

4.3 Comparative Protocol

For each dataset and applicable cartogram type, the same input polygons, region values and with that target areas were supplied to the framework and to the corresponding baseline. All metrics defined in Section 4.4 were then computed from the resulting output polygons, independently of which system produced them, using a single evaluation module, so that no metric implementation detail differs between the two systems being compared. Runtime was measured separately from data loading to final polygon output.

4.4 Evaluation Metrics

This thesis adopts the metrics defined by Alam, Kobourov and Veeramoni [9] and extends it with two metrics that are specific to the optimization-based setting of this framework and derived from common errors occurred during development: an overlap/topological-integrity check and a gap-size metric, both formalized here using Shapely predicates since no standard definition exists in the literature for optimization-based (as opposed to physically-simulated) cartogram construction.

4.4.1 Statistical Accuracy: Cartographic Error

The most basic requirement of a cartogram is that region areas match their target values. Following Alam et al. [9], the cartographic error for region $\mathcal{R}_i$ is defined as the relative discrepancy between the measured area A_i and the target area A_i^* . Let A_i denote the Shoelace area of region i ’s output polygon:

$$\epsilon_v = \frac{|A_i - A_i^*|}{\max(A_i, A_i^*)} \quad (4.1)$$

The framework reports both the mean error $\epsilon = \frac{1}{N} \sum_i \epsilon_i$ and the maximum error $\xi = \max_i \epsilon_i$ over all N regions, since a low mean can mask a small number of severely under- or over-scaled regions. In addition, one complementary area measure is reported:

$$\text{SR}(N) = \frac{|A_i - A_i^{(0)}|}{|A_i^* - A_i^{(0)}|} \quad (4.2)$$

where $A_i^{(0)}$ is the region’s original (pre-transformation) area. $\text{SR}(N)$ computes how much of an intended change was achieved.

4.4.2 Geographical Accuracy: Shape Preservation

Hamming distance [53] : Following the shape-distortion measure of Alam, Kobourov and Veeramoni [9], each region polygon is centered on its own centroid and rescaled to unit area in the map and, separately, in the cartogram, and the two normalized shapes are superimposed. The Hamming distance is then half the area of their symmetric difference,

$$\delta_v = \frac{1}{2} \text{area}(P_v^{\text{map}} \triangle P_v^{\text{cartogram}}), \quad \delta_v \in [0, 1], \quad (4.3)$$

with $\delta_v = 0$ for a perfectly preserved shape and $\delta_v \rightarrow 1$ for shapes with no overlap after normalization.

4.4.3 Geographical Accuracy: Positional Fidelity

Angular orientation error. Following Heilmann et al. [28], for every pair of regions (i, j) the angle of the line connecting their centroids is compared before and after transformation:

$$\theta_{ij} = |\angle(c_i, c_j)_{\text{orig}} - \angle(c_i, c_j)_{\text{new}}|, \quad (4.4)$$

taken modulo 2π and reduced to $[0, \pi]$, then averaged over all pairs.

Orthogonal (relative-direction) error. A coarser but more interpretable companion metric checks, for each pair of regions, whether their relative north-south and east-west ordering is preserved:

$$\rho = \frac{1}{\binom{n}{2}} \sum_{i < j} \mathbb{I}[\text{sgn}(\Delta x)_{\text{orig}} \neq \text{sgn}(\Delta x)_{\text{new}} \vee \text{sgn}(\Delta y)_{\text{orig}} \neq \text{sgn}(\Delta y)_{\text{new}}] \quad (4.5)$$

A cartogram with low θ but non-zero ρ indicates that some region has crossed from one cardinal side of a neighbor to another, a qualitatively worse error for readability than a small continuous angular drift.

4.4.4 Topological Accuracy: Adjacency Error

Topological accuracy asks whether the cartogram preserves which regions are neighbors, independent of shape or position. An adjacency graph E is built for the original map and for the cartogram output, using the same vertex-sharing test the solver itself uses to construct T (Pre-processing, tolerance τ_{adj}), i.e. τ measures adjacency preservation with respect to exactly the neighbour relation the framework optimizes against, not a separately-defined geometric test. The adjacency error is then the graph dissimilarity

$$\tau = 1 - \frac{|E_{\text{cart}} \cap E_{\text{orig}}|}{|E_{\text{cart}} \cup E_{\text{orig}}|}. \quad (4.6)$$

For the contiguous mode, τ should be going towards zero by construction, since the framework tries to optimize towards shared-vertex equality between adjacent regions.

4.4.5 Topological Integrity: Overlap and Self-Intersection

Because the framework solves a continuous optimization problem rather than an explicitly overlap-free physical simulation, region overlap is a distinct failure mode that the metrics above do not capture, two polygons can have perfect area, shape, and adjacency scores while partially occupying the same space. This thesis therefore introduces a direct topological-integrity check, computed with Shapely: for every pair of regions, the geometries are validity-repaired and their pairwise intersection area is accumulated. The reported quantities are the count of self-intersecting (invalid) polygons, the count of pairwise overlaps, and the overlap area as a fraction of total map area:

$$\Phi_{\text{overlap}} = \frac{\sum_{i < j} \text{area}(P'_i \cap P'_j)}{\sum_i A_i^{(0)}}. \quad (4.7)$$

4.4.6 Topological Integrity: Gap Error

The converse failure mode, previously adjacent regions pulling apart and leaving a visible gap, is not captured by area, shape, adjacency, or overlap metrics either. For every pair of regions adjacent in the original map (using the polygon-intersection adjacency test $T_\cap$ defined in Section 4.4.4, not the solver’s vertex-sharing set T), the gap contribution is the excess area of the convex hull of that pair’s cartogram output over the sum of their individual areas:

$$\Phi_{\text{gap}} = \frac{1}{\sum_i A_i^{(0)}} \sum_{(i,j) \in T_\cap} \max\left(0, \text{area}(\text{hull}(P'_i \cup P'_j)) - \text{area}(P'_i) - \text{area}(P'_j)\right), \quad (4.8)$$

normalized by the total original map area. Because each region can contribute to several pairwise terms (once per originally-adjacent neighbour), a gap between two regions with many mutual neighbours can be counted more than once. Unlike Φ_{overlap} , Φ_{gap} is therefore not bounded by 1, and larger values should be read as ”more total gap area, possibly double-counted across shared neighbourhoods” rather than as a literal fraction of the map.

4.4.7 Complexity and Runtime

Finally, two practical axes are reported that do not concern quality directly but matter for the framework’s stated goal of a single, general-purpose solver: the mean and maximum polygon vertex count per region as complexity metrics, and runtime, reported separately per cartogram type and dataset size.

4.5 Results

4.5.1 Computing Time

Table 4.1 reports runtime for every cartogram type and dataset, pairing each baseline directly against the corresponding Framework (CF) output. Contiguous mode is compared against two baselines (Gastner cpp and R). Demers has no baseline, since Section 4.1 noted none was available.

Dataset	Type	Contiguous		Dorling		Non-contiguous		Demers
		Baseline	Framework	Baseline (R)	Framework	Baseline (R)	Framework	
USA	Gastner cpp	1,082ms	194,400ms	90ms	32,600ms	410ms	33,800ms	19,100ms
	R (conti)	11,230ms						
India	Gastner cpp	958ms	203,600ms	10ms	12,200ms	190ms	13,900ms	8,500ms
	R (conti)	1,450ms						
Taiwan & China	Gastner cpp	966ms	220,500ms	30ms	9,700ms	260ms	8,700ms	7,400ms
	R (conti)	1,600ms						
World	Gastner cpp	60,113ms	1,611,600ms	440ms	1,747,000ms	1,310ms	125,500ms	293,000ms
	R (conti)	16,130ms						
Netherlands	Gastner cpp	119ms	41,100ms	40ms	1,100ms	230ms	500ms	700ms
	R (conti)	1,530ms						
Germany	Gastner cpp	409ms	61,200ms	60ms	1,800ms	270ms	1,800ms	1,300ms
	R (conti)	1,510ms						

Table 4.1: Runtime per cartogram type and dataset, with baselines and Framework (CF) output paired for direct comparison. Contiguous mode has two baselines (Gastner cpp, R). Demers has no baseline.

4.5.2 Per-Dataset Comparison

In this section the results of the experiments are shown. Dataset wise with all eight visualized outputs of the baseline and the Framework as well as computed quality metrics. All metrics are reported to the third value behind the point if applicable. Exact Values are derived from exact results other numbers of values behind the point are derived from rounding. A standard map would have a high $\bar{\epsilon}/\xi$: mean/max cartographic error 4.1, a perfect zero $\bar{\delta}$: mean Hamming distance 4.3 a perfect zero τ : adjacency error 4.6, and also no Φ_{overlap} : overlap area fraction 4.7

USA

Figure 4.1 shows the USA dataset (electoral college votes per state) under all four baselines and all four Framework modes. Table 4.2 reports the corresponding statistical and topological accuracy metrics.

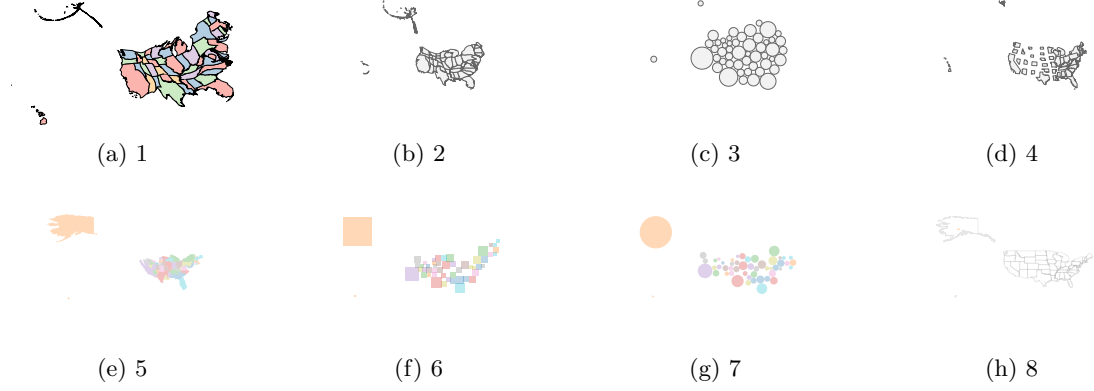


Figure 4.1: USA: 1. Gastner cpp; 2. R-conti; 3. R-dorling; 4. R-noncontiguous; 5. Framework contiguous; 6. Framework demers; 7. Framework dorling; 8. Framework non-contiguous.

System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}
<i>Contiguous</i>					
Gastner cpp	0.016	0.242	0.326	0.045	0
R (conti)	0.052	0.964	0.273	0	0.000
Framework	0.446	0.989	0.248	0.244	0
<i>Dorling</i>					
R (dorling)	0.000	0.000	0.257	1	0
Framework	0.000	0.000	0.257	1	0.000
<i>Non-contiguous</i>					
R (non)	0.012	0.113	0.181	1	0.097
Framework	0.001	0.012	0.000	1	0.000
<i>Demers</i>					
Framework	0.000	0.000	0.269	0.768	0.000

Table 4.2: Statistical and topological accuracy for the USA dataset. $\bar{\epsilon}/\xi$: mean/max cartographic error 4.1; $\bar{\delta}$: mean Hamming distance 4.3; τ : adjacency error 4.6; Φ_{overlap} : overlap area fraction 4.7. Full metric set (including E_{max} , $SR(N)$, angular/orthogonal position error, gap fraction, vertex counts) is reported per-dataset in Table 7.1.

India

Figure 4.2 shows the India dataset (gdp per state) under all four baselines and all four Framework modes. Table 4.3 reports the corresponding statistical and topological accuracy metrics.

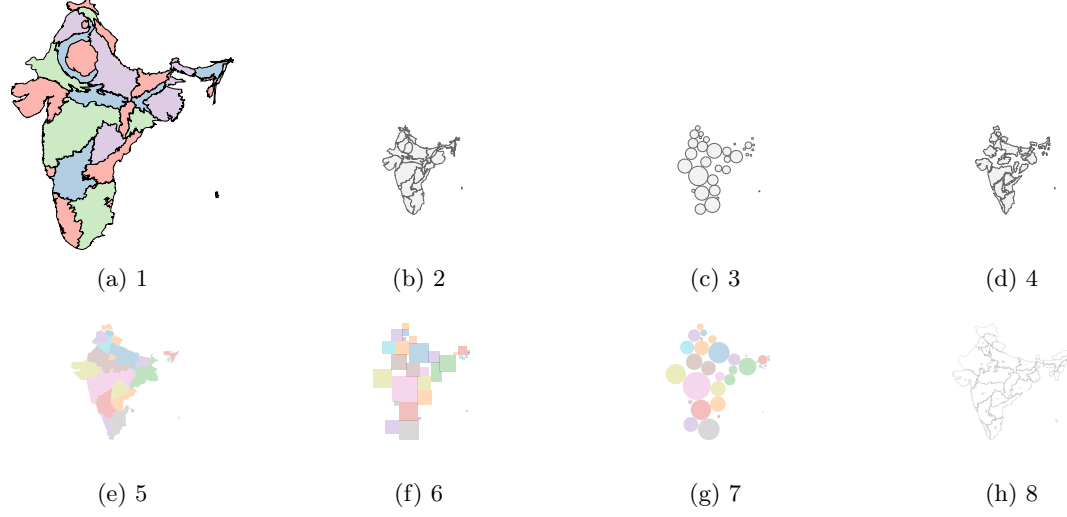


Figure 4.2: India. 1. Gastner cpp; 2. R-conti; 3. R-dorling; 4. R-noncontiguous; 5. Framework contiguous; 6. Framework demers; 7. Framework dorling; 8. Framework noncontiguous.

System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}
<i>Contiguous</i>					
Gastner cpp	0.002	0.007	0.296	0	0.000
R (conti)	0.107	0.759	0.296	0	0.000
Framework	0.339	0.988	0.198	0.262	0.000
<i>Dorling</i>					
R (dorling)	0.071	0.147	0.263	1	0
Framework	0.000	0.000	0.267	1	0
<i>Non-contiguous</i>					
R (non)	0.071	0.143	0.030	1	0.068
Framework	0.000	0.001	0.000	1	0
<i>Demers</i>					
Framework	0.000	0.000	0.283	0.721	0.000

Table 4.3: Statistical and topological accuracy for the India dataset. $\bar{\epsilon}/\xi$: mean/max cartographic error 4.1; $\bar{\delta}$: mean Hamming distance 4.3; τ : adjacency error 4.6; Φ_{overlap} : overlap area fraction 4.7. Full metric set (including E_{max} , $SR(N)$, angular/orthogonal position error, gap fraction, vertex counts) is reported per-dataset in Table 7.1.

Taiwan & China

Figure 4.3 shows the Taiwan & China dataset (gdp per state) under all four baselines and all four Framework modes. Table 4.4 reports the corresponding statistical and topological accuracy metrics.

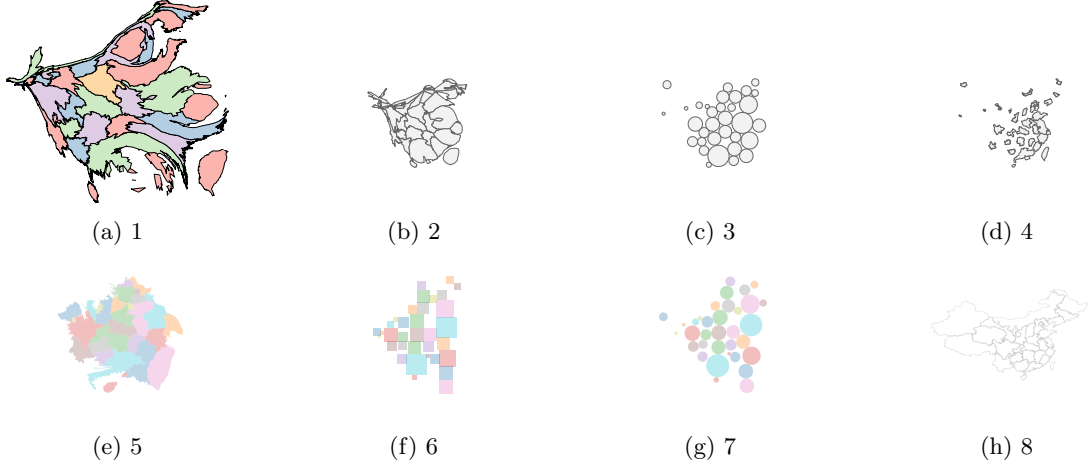


Figure 4.3: Taiwan & China. 1. Gastner cpp; 2. R-conti; 3. R-dorling; 4. R-noncontiguous; 5. Framework contiguous; 6. Framework demers; 7. Framework dorling; 8. Framework noncontiguous.

System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}
<i>Contiguous</i>					
Gastner cpp	0.034	0.442	0.381	0	0
R (conti)	0.204	0.904	0.370	0.879	0.000
Framework	0.522	0.966	0.24	0.398	0
<i>Dorling</i>					
R (dorling)	0.088	0.213	0.254	1	0
Framework	0.000	0.000	0.261	1	0
<i>Non-contiguous</i>					
R (non)	0.075	0.213	0.039	1	0.087
Framework	0.014	0.337	0.000	1	0
<i>Demers</i>					
Framework	0.000	0.000	0.274	0.797	0

Table 4.4: Statistical and topological accuracy for the Taiwan & China dataset. $\bar{\epsilon}/\xi$: mean/max cartographic error 4.1; $\bar{\delta}$: mean Hamming distance 4.3; τ : adjacency error 4.6; Φ_{overlap} : overlap area fraction 4.7. Full metric set (including E_{max} , $SR(N)$, angular/orthogonal position error, gap fraction, vertex counts) is reported per-dataset in Table 7.1.

Netherlands

Figure 4.4 shows the Netherlands dataset (population per state) under all four baselines and all four Framework modes. Table 4.5 reports the corresponding statistical and topological accuracy metrics.

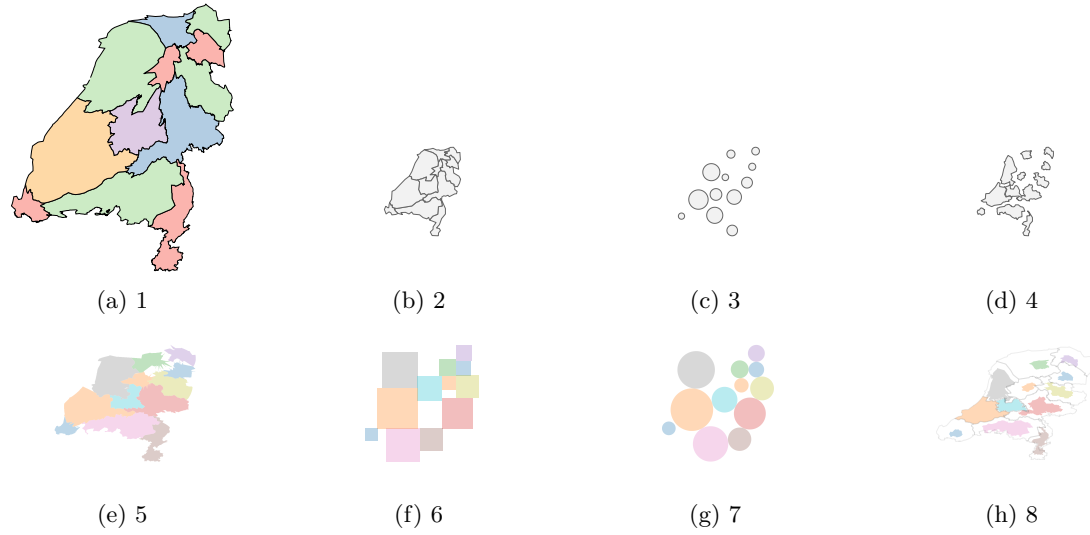


Figure 4.4: Netherlands. 1. Gastner cpp; 2. R-conti; 3. R-dorling; 4. R-noncontiguous; 5. Framework contiguous; 6. Framework demers; 7. Framework dorling; 8. Framework noncontiguous.

System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}
<i>Contiguous</i>					
Gastner cpp	0.001	0.004	0.272	0	0.000
R (conti)	0.031	0.126	0.189	0	0.000
Framework	0.133	0.393	0.164	0.217	0
<i>Dorling</i>					
R (dorling)	0.012	0.025	0.230	1	0
Framework	0.000	0.000	0.285	1	0
<i>Non-contiguous</i>					
R (non)	0.012	0.025	0.033	1	0.002
Framework	0.000	0.000	0.050	1	0.003
<i>Demers</i>					
Framework	0.000	0.000	0.452	0.739	0.000

Table 4.5: Statistical and topological accuracy for the Netherlands dataset. $\bar{\epsilon}/\xi$: mean/max cartographic error 4.1; $\bar{\delta}$: mean Hamming distance 4.3; τ : adjacency error 4.6; Φ_{overlap} : overlap area fraction 4.7. Full metric set (including E_{max} , $SR(N)$, angular/orthogonal position error, gap fraction, vertex counts) is reported per-dataset in Table 7.1.

Germany

Figure 4.5 shows the Germany dataset (population per state) under all four baselines and all four Framework modes. Table 4.6 reports the corresponding statistical and topological accuracy metrics.

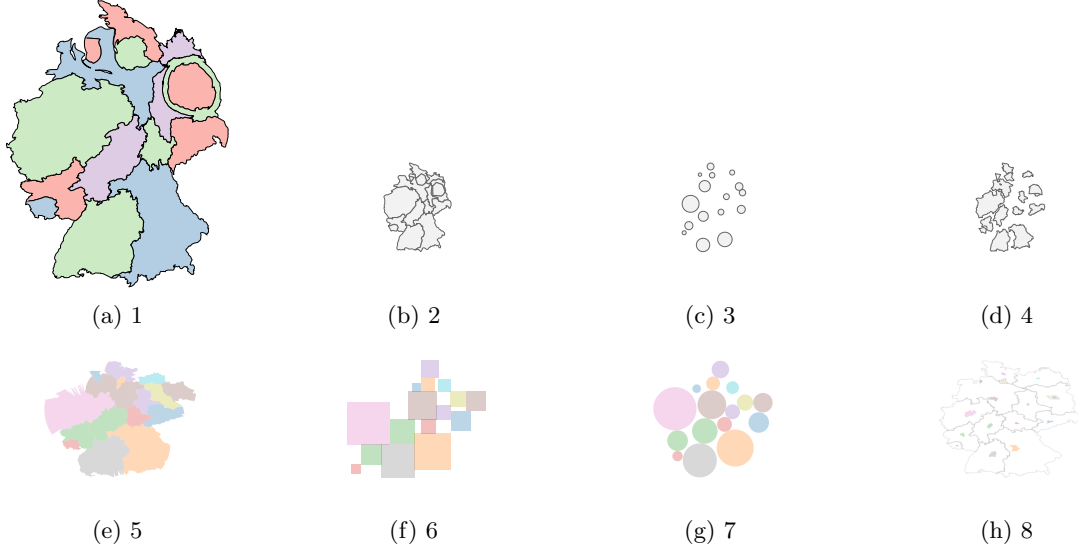


Figure 4.5: Germany. 1. Gastner cpp; 2. R-conti; 3. R-dorling; 4. R-noncontiguous; 5. Framework contiguous; 6. Framework demers; 7. Framework dorling; 8. Framework noncontiguous.

System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}
<i>Contiguous</i>					
Gastner cpp	0.076	0.572	0.243	0	0.042
R (conti)	0.124	0.581	0.199	0	0.034
Framework	0.161	0.781	0.145	0.1	0
<i>Dorling</i>					
R (dorling)	0.033	0.07	0.211	1	0
Framework	0.000	0.000	0.247	1	0
<i>Non-contiguous</i>					
R (non)	0.035	0.068	0.012	1	0.031
Framework	0.000	0.000	0.000	1	0
<i>Demers</i>					
Framework	0.000	0.000	0.263	0.759	0.000

Table 4.6: Statistical and topological accuracy for the Germany dataset. $\bar{\epsilon}/\xi$: mean/max cartographic error 4.1; $\bar{\delta}$: mean Hamming distance 4.3; τ : adjacency error 4.6; Φ_{overlap} : overlap area fraction 4.7. Full metric set (including E_{max} , $SR(N)$, angular/orthogonal position error, gap fraction, vertex counts) is reported per-dataset in Table 7.1.

World

Figure 4.6 shows the World dataset (population per state) under all four baselines and all four Framework modes. Table 4.7 reports the corresponding statistical and topological accuracy metrics.

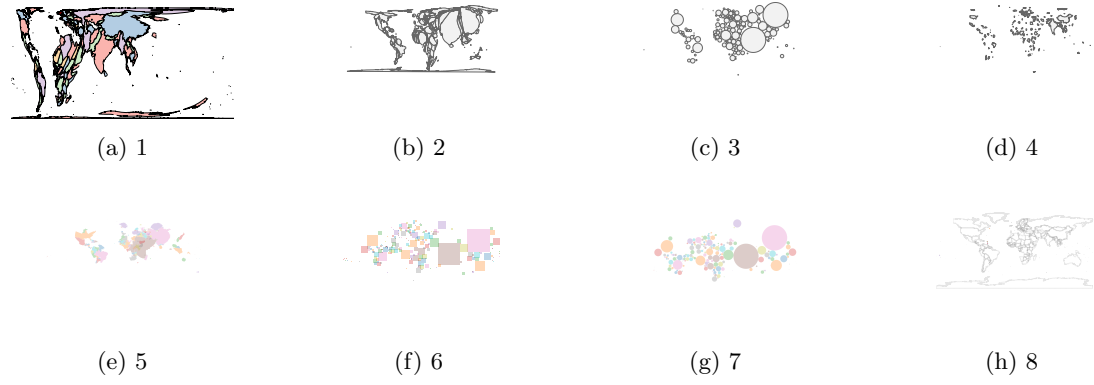


Figure 4.6: World. 1. Gastner cpp; 2. R-conti; 3. R-dorling; 4. R-noncontiguous; 5. Framework contiguous; 6. Framework demers; 7. Framework dorling; 8. Framework noncontiguous.

System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}
<i>Contiguous</i>					
Gastner cpp	0.134	1	0.491	0	0.007
R (conti)	0.333	1	0.394	0.007	0.000
Framework	0.556	0.999	0.25	0.694	0.039
<i>Dorling</i>					
R (dorling)	0.148	0.711	0.313	1	0
Framework	0.012	1	0.301	1	0.000
<i>Non-contiguous</i>					
R (non)	0.129	0.728	0.037	1	0.01
Framework	0.106	1	0.000	1	0
<i>Demers</i>					
Framework	0.000	0.000	0.310	0.763	0.000

Table 4.7: Statistical and topological accuracy for the World dataset. $\bar{\epsilon}/\xi$: mean/max cartographic error 4.1; $\bar{\delta}$: mean Hamming distance 4.3; τ : adjacency error 4.6; Φ_{overlap} : overlap area fraction 4.7. Full metric set (including E_{max} , $SR(N)$, angular/orthogonal position error, gap fraction, vertex counts) is reported per-dataset in Table 7.1.

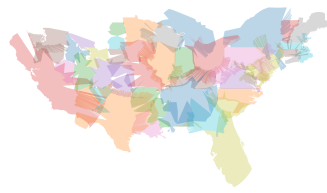
4.5.3 Postprocessing

The following figures and tables compare Contiguous and Dorling output before and after the post-processing pipeline of Section 3.12, for each dataset in turn. Non-contiguous and Demers modes are omitted here since post-processing does not alter their output (Section 3.12).

USA

System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}	Gap
<i>Contiguous</i>						
Pre-postprocessing	0.32	0.975	0.245	1	0.24	1.546
Post-postprocessing	0.446	0.989	0.248	0.244	0	1.212
<i>Dorling</i>						
Pre-postprocessing	0.000	0.000	0.257	1	0.028	1.921
Post-postprocessing	0.000	0.000	0.257	1	0	2.094

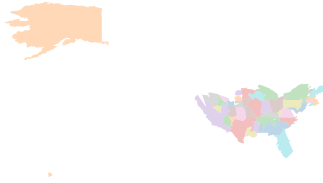
Table 4.8: Quality Criteria of the Dorling and Contiguous Cartograms of the Cartogram Framework pre and post postprocessing for the USA.



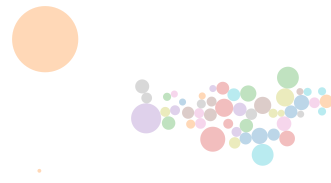
(a) 1



(b) 2



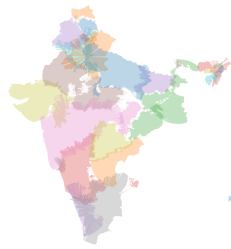
(c) 3



(d) 4

Figure 4.7: USA: 1. Contiguous pre post-processing; 2. Dorling pre post-processing; 3. Contiguous post post-processing; 4. Dorling post post-processing;

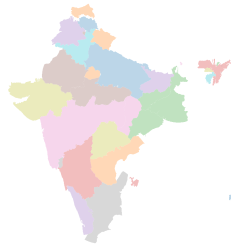
India



(a) 1



(b) 2



(c) 3



(d) 4

Figure 4.8: India: 1. Contiguous pre post-processing; 2. Dorling pre post-processing; 3. Contiguous post post-processing; 4. Dorling post post-processing;

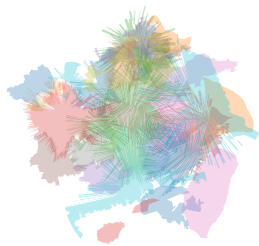
System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}	Gap
<i>Contiguous</i>						
Pre-postprocessing	0.057	0.305	0.11	1.0	0.224	2.193
Post-postprocessing	0.339	0.988	0.198	0.262	0.000	1.839
<i>Dorling</i>						
Pre-postprocessing	0.000	0.000	0.267	1.0	0.021	1.026
Post-postprocessing	0.000	0.000	0.267	1.0	0.0	1.142

Table 4.9: Quality Criteria of the Dorling and Contiguous Cartograms of the Cartogram Framework pre and post postprocessing for India.

Taiwan & China

System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}	Gap
<i>Contiguous</i>						
Pre-postprocessing	0.182	0.908	0.235	1	0.639	2.233
Post-postprocessing	0.522	0.966	0.24	0.398	0	1.063
<i>Dorling</i>						
Pre-postprocessing	0.000	0.000	0.261	1	0.032	1.668
Post-postprocessing	0.000	0.000	0.261	1	0	1.836

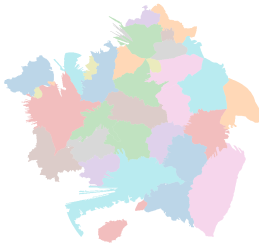
Table 4.10: Quality Criteria of the Dorling and Contiguous Cartograms of the Cartogram Framework pre and post postprocessing for Taiwan & China.



(a) 1



(b) 2



(c) 3



(d) 4

Figure 4.9: Taiwan & China: 1. Contiguous pre post-processing; 2. Dorling pre post-processing; 3. Contiguous post post-processing; 4. Dorling post post-processing;

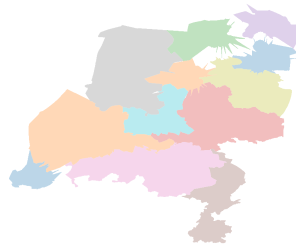
Netherlands



(a) 1



(b) 2



(c) 3



(d) 4

Figure 4.10: Netherlands: 1. Contiguous pre post-processing; 2. Dorling pre post-processing; 3. Contiguous post post-processing; 4. Dorling post post-processing;

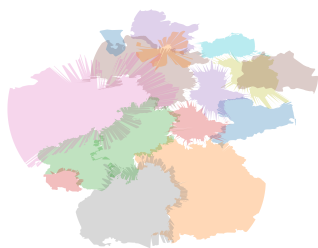
System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}	Gap
<i>Contiguous</i>						
Pre-postprocessing	0.120	0.287	0.128	1	0.123	1.851
Post-postprocessing	0.133	0.393	0.164	0.217	0	1.679
<i>Dorling</i>						
Pre-postprocessing	0.000	0.000	0.285	1	0.024	0.928
Post-postprocessing	0.000	0.000	0.285	1	0	1.043

Table 4.11: Quality Criteria of the Dorling and Contiguous Cartograms of the Cartogram Framework pre and post postprocessing for Netherlands.

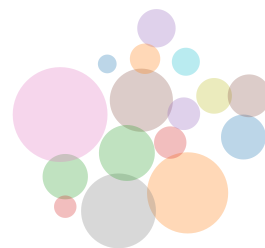
Germany

System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}	Gap
<i>Contiguous</i>						
Pre-postprocessing	0.055	0.128	0.102	1	0.136	1.562
Post-postprocessing	0.161	0.781	0.145	0.1	0	1.321
<i>Dorling</i>						
Pre-postprocessing	0.000	0.000	0.247	1	0.024	0.685
Post-postprocessing	0.000	0.000	0.247	1	0	0.799

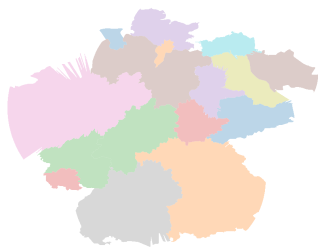
Table 4.12: Quality Criteria of the Dorling and Contiguous Cartograms of the Cartogram Framework pre and post postprocessing for Germany.



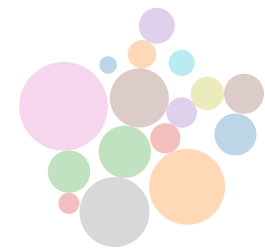
(a) 1



(b) 2



(c) 3



(d) 4

Figure 4.11: Germany: 1. Contiguous pre post-processing; 2. Dorling pre post-processing; 3. Contiguous post post-processing; 4. Dorling post post-processing;

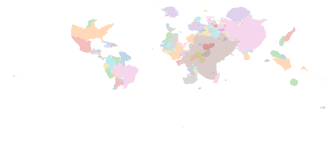
World



(a) 1



(b) 2



(c) 3



(d) 4

Figure 4.12: World: 1. Contiguous pre post-processing; 2. Dorling pre post-processing; 3. Contiguous post post-processing; 4. Dorling post post-processing;

System	$\bar{\epsilon}$	ξ	$\bar{\delta}$ (Hamming)	τ (adjacency)	Φ_{overlap}	Gap
<i>Contiguous</i>						
Pre-postprocessing	0.263	0.997	0.173	0.997	0.588	4.086
Post-postprocessing	0.556	0.999	0.25	0.694	0.039	1.135
<i>Dorling</i>						
Pre-postprocessing	0.012	1	0.301	1	0.020	2.381
Post-postprocessing	0.012	1	0.301	1	0.000	2.439

Table 4.13: Quality Criteria of the Dorling and Contiguous Cartograms of the Cartogram Framework pre and post postprocessing for World.

Chapter 5

Discussion

This chapter discusses the results of this Thesis generated in Chapter 4.5 with taking the five research questions raised in Chapter 1 into account.

The Framework manages to generate three of the four Cartogram types in a comparable manner to specialized baseline implementations. Non-contiguous and Demers without any postprocessing and Dorling with postprocessing can challenge the results of the baseline implementations. Dorling without post-processing is left with slight overlap due to its circle shape. For Non-contiguous cartograms, the framework achieves a mean cartographic error lower than R's Olson-based implementation on every dataset (e.g. USA: $\bar{\epsilon} = 0.001$ vs. 0.012, Taiwan & China: 0.014 vs. 0.075), while its Hamming distance remains smaller than 0.0005. The difference between the baseline and the Framework could stem from the different approaches to combine readability, low cartographic error and non overlap. If every region gets scaled according to the target value while the overall area is kept this almost inevitably leads to overlap. While the R baseline seems to have chosen an approach where small cartographic error was allowed the framework tried to keep that as low as possible leading to shrinking most of the regions. The Dorling cartogram achieves consistently on five out of the six datasets a negligible small cartographic error smaller than 0.0005 which the baseline only achieves on the usa dataset. For the world dataset the cartographic error on Dorling rises to 0.012 but still stays by over the factor ten smaller than the cartographic error on the baseline. The Hamming distance for the Dorlings cartograms is by construction in both the baseline and the framework case very similar since the regions get deformed to circles both times. The Demers cartogram doesn't have a baseline to be compared to, but the results fit the expectations of negligible cartographic error good to perfect adjacencies and no overlap. These results show that a single mathematical optimization space can describe three different cartograms. An exception here is the contiguous cartogram with mean cartographic errors of 13–56% across the six datasets, against 0.1–13% for gastners and 3–33% for the R baseline. Even the worst case of Gastner's cpp roughly matches the frameworks best mean cartographic error while also the R baseline stays consistently under the Framework.

While the contiguous cartogram has bad results regarding the cartographic error, its shape preservation is uniformly better than the baselines across datasets. For all datasets, the framework achieves a lower Hamming distance than both baselines (USA: 0.242 vs. Gastner's 0.273 and R's 0.326, India : 0.198 vs. Gastner's 0.296 and R's 0.296, Taiwan & China: 0.24 vs. Gastner's 0.381 and R's 0.370, Netherlands: 0.164 vs. Gastner's 0.272 and R's 0.189, Germany: 0.145 vs. Gastner's 0.243 and R's 0.199 and the world: 0.25 vs. Gastner's 0.491 and R's 0.394).

RQ1 asked for a fast framework. As it can be seen in Table 4.1, this was not achieved. Every mode of the framework is slower than its corresponding baseline. This can partly be explained by the size of the combined optimization problem, and partly by the fact that the framework is implemented in Python rather than a compiled language, leaving room for future performance work.

Post-processing (Section 3.12) has different effects on contiguous and Dorling output. For contiguous cartograms, it substantially worsens cartographic error on every dataset (USA: $\bar{\epsilon} = 0.32 \rightarrow 0.446$, India: $0.057 \rightarrow 0.339$, Taiwan & China: $0.182 \rightarrow 0.522$, Netherlands: $0.120 \rightarrow 0.133$, Germany: $0.055 \rightarrow 0.161$ and the World: $0.263 \rightarrow 0.556$) while reducing the adjacency error τ (USA: $1.0 \rightarrow 0.244$; India: $1.0 \rightarrow 0.262$; Taiwan & China: $1.0 \rightarrow 0.398$, Netherlands: $1 \rightarrow 0.217$, Germany: $1 \rightarrow 0.1$ and the World: $0.997 \rightarrow 0.694$) and eliminating overlap entirely on all six. The mechanism is direct: disputed-pixel are reassigned to close gaps and remove overlap, which necessarily perturbs each region’s measured area away from whatever the QP had already achieved. The post-processing stage is trading the area accuracy the solver achieved for topological accuracy. This is consistent with the vertex-penetration limitation identified in Section 2.3: the QP alone cannot guarantee non-overlap, so closing the remaining gaps geometrically is unavoidable if a usable (non-self-intersecting, adjacency-preserving) cartogram is the goal, but it comes at a cost to the other quality criteria.

Dorling cartograms show a different pattern: post-processing (circle declumping, Section 3.12) leaves cartographic error, Hamming distance and adjacency completely unchanged on every dataset, since the relaxation only ever moves circle centres and never changes their radii or moves vertices so that area accuracy is immune to this stage. What it does change, is overlap, which drops to zero on all six datasets (USA: $0.028 \rightarrow 0$, India: $0.021 \rightarrow 0$, Taiwan & China: $0.032 \rightarrow 0$, Netherlands: $0.024 \rightarrow 0$, Germany: $0.024 \rightarrow 0$ and the World: $0.02 \rightarrow 0$), at the cost of a small increase in the gap metric (USA: $1.921 \rightarrow 2.094$, India: $1.026 \rightarrow 1.142$, Taiwan & China: $1.668 \rightarrow 1.836$, Netherlands: $0.928 \rightarrow 1.043$, Germany: $0.685 \rightarrow 0.799$ and the World: $2.381 \rightarrow 2.439$) since pushing circles apart to remove overlap necessarily increases the empty space between them. Unlike the contiguous case, this is a clean trade with no cost to the framework’s primary quality criterion, which is why Dorling was one of the three modes described in this chapter as competitive with, or better than, its specialized baseline.

Returning to the five research questions posed in Chapter 1: RQ1 asked whether the framework could be fast, and Table 4.1 answers this directly: no, every mode is substantially slower than its specialized counterpart. RQ2 and RQ3 asked which quality criteria a unified framework needs to encode and how they can be formalized within a single optimization model. The mapping in Table 3.1 motivated encoding cartographic error, shape deformation, spatial deformation and topological accuracy directly as QP terms (Sections 3.10–3.11), and the results in this chapter show this choice was sufficient for Demers, Dorling and non-contiguous cartograms, which reach quality competitive with or better than their specialized baselines. For contiguous cartograms, the same terms are necessary but not sufficient: without a convex non-intersection constraint, adjacency and shape can only be encoded softly, leaving an irreducible gap in cartographic error relative to Gastner’s cpp and R (Section 2.3). This also exposes a trade-off the mapping in Table 3.1 does not make explicit on its own: strengthening topological accuracy, whether through $\lambda_t/\lambda_{\text{con}}$ or through the post-processing pass, comes at a direct cost to cartographic error for contiguous cartograms, whereas Demers, Dorling and non-contiguous modes show no comparable conflict, since their target shapes leave the topology terms largely inactive or already satisfied. RQ4, on how the framework compares to specialized algorithms, is answered in detail across Section 4.5: it matches or exceeds baseline quality on three of the four types but is slower

across all of them, so unification trades runtime broadly, not only contiguous-mode quality, for generality. RQ5 asked whether an automated recommender could be built on top of the computed quality metrics. Section 3.13 designs and implements such a system, mapping six interpretable sliders onto the framework’s native parameters, but its recommendations were not evaluated against user judgments or ground-truth type suitability in this thesis. RQ5 is therefore answered only at the design level here; its empirical validation is left as a limitation (Section 6.1) and a direction for future work (Section 6.2).

Both post-processing raise a question in the framework’s central goal. The elegance of a single convex program is that every cartogram type is an optimum of one problem, with no style-specific code. But for contiguous cartograms in particular, a genuinely usable output is only reached by leaving that optimum and applying a separate, post-processing step, one that is not itself part of the optimization problem.

Chapter 6

Conclusion

This thesis set out to test whether a single convex quadratic program could unify cartogram generation across multiple styles, rather than requiring a separate, isolated algorithm per style. With Demers, Dorling and non-contiguous cartograms being reproduced at quality competitive with and on several metrics better than, specialized reference implementations this can be answered positively, confirming that these three styles really are optima of one shared formulation rather than requiring bespoke algorithms.

Contiguous cartograms are the exception: the framework produces usable, but measurably worse, output than both baselines, and the adjacency-error results in Chapter 4.5 pin the cause down to the same non-convexity of polygon non-intersection identified in Section 2.3 the framework does not solve that open problem, but it does show precisely how the soft workaround around it trades area accuracy and adjacency for feasibility.

The cost of unification is speed: every mode is substantially slower than its specialized counterpart, not only the contiguous case, which is a genuine limitation of building one general solver rather than several specialized ones.

6.1 Limitations

The results of this thesis should be interpreted in light of several limitations. First, the framework demonstrates what can be achieved within a single mathematical formulation, rather than establishing the full range of possibilities achievable by specialised algorithms. A specialised method can exploit assumptions and structures that are specific to one cartogram style, whereas the proposed framework deliberately sacrifices some of this specialisation in favour of a common representation. Consequently, the comparison is not intended to show that the unified approach is universally superior, but rather that a substantial degree of unification is possible without making the resulting cartograms unusable.

A particularly important limitation concerns vertex penetration in contiguous mode. The homothetic vertex transformation allows individual vertices to move radially from their original positions as the corresponding region is enlarged or reduced. When neighbouring regions have substantially different target areas, these transformations can cause vertices to penetrate into adjacent polygons. Preventing this behaviour through a hard constraint is difficult within a convex optimisation framework. The feasible region corresponding to non-intersecting polygons

with movable vertices is generally non-convex, while a natural half-plane separation formulation introduces products between optimisation variables and therefore breaks the convexity of the problem. During development, hard versions of some of the objectives were tested unfortunately leading to infeasibilities in the interaction with each other.

Not all of the metrics newly introduced in the Section 4.4 proved equally informative in practice. The overlap fraction turned out to be one of the more useful additions of this thesis: it isolated a failure mode that area, shape and adjacency scores could not detect on their own, and it tracked cleanly with the post-processing effects discussed in Chapter 4.5. Success Rate was added with the idea to give another angle towards the cartographic error but ended up not giving valuable information. The gap metric was similarly limited, its absolute values were hard to interpret across datasets of different density, and it mainly confirmed, rather than added to, what the overlap and adjacency metrics already showed in a clearer way.

A second limitation is the substantial runtime overhead observed across all cartogram modes. The proposed framework is consistently slower than the specialised reference implementations. This is not limited to the contiguous case and is therefore not solely a consequence of the difficulty of enforcing polygon non-intersection. Rather, it is a direct consequence of solving a larger and more general constraint system for every instance. While this cost is acceptable for demonstrating the feasibility of a unified approach, it limits the immediate applicability of the framework to settings where large numbers of cartograms must be generated quickly.

Finally, no direct baseline comparison was available for the Demers cartograms. No maintained, publicly available implementation of the Nickel/Sondag/Meulemans-style LP-based Demers cartogram could be identified for a direct comparison. The Demers results are therefore reported as standalone results rather than against a specialised reference implementation. This means that the conclusions concerning the quality of the Demers formulation are necessarily more limited than those for the styles for which direct baseline comparisons were available.

6.2 Future Work

Several directions follow naturally from the limitations and results of this thesis. One promising extension would be to increase the degrees of freedom available to each vertex to address the problem of hard vertex locking turning infeasible. The current formulation primarily allows a vertex to move along the axis defined by its original position relative to the centre of the region. An alternative would be to allow an additional degree of freedom 90° to this axis, giving each vertex greater flexibility while retaining a relatively structured transformation. A preliminary version of this approach was tested during the development of the framework, but it was not investigated in depth because the initial experiments did not produce immediate improvements. A more systematic exploration, potentially combined with different regularisation terms, could determine whether the additional freedom improves area accuracy or reduces distortion.

Finally, the unified formulation creates opportunities to explore cartogram styles that are not directly targeted by the current framework. Because the underlying optimisation model separates general geometric objectives from style-specific transformations and constraints, it may be possible to construct hybrid cartogram types by combining components of the existing modes. Such hybrid formulations could provide a useful test of the central premise of this thesis: that cartogram styles can be viewed not only as separate algorithms, but as different points in a broader design space defined by a common set of geometric objectives and constraints. More directed could this also be used to implement the features necessary to generate the other defined

types such as Mosaic cartograms.

Overall, the results demonstrate that a single convex quadratic programming framework can capture a broad range of cartogram styles. The approach does not eliminate the need for specialised methods, particularly where strict polygon non-intersection or high computational efficiency is required. Nevertheless, the ability to reproduce several distinct cartogram styles within one optimisation framework suggests that unification is a viable direction for cartogram generation and provides a foundation for further exploration of both existing and hybrid cartogram types.

Bibliography

- [1] T. Munzner, *Visualization analysis & design*. Boca Raton, FL: CRC Press/Taylor & Francis Group, 2015. [Online]. Available: <https://doi.org/10.1201/b17511>
- [2] S. Nusrat and S. Kobourov, “The state of the art in cartograms,” *Computer Graphics Forum*, vol. 35, no. 3, pp. 619–642, 2016. [Online]. Available: <https://doi.org/10.1111/cgf.12932>
- [3] H. W. Holbrook and C. Cartwright, “A methodology for conducting a study into the communicative effectiveness between choropleth maps and cartograms,” *Internal Report, IR/16/021*, 2016.
- [4] S. Kaspar, S. I. Fabrikant, and P. Freckmann, “Empirical study of cartograms,” in *25th international cartographic conference*, vol. 3, 2011, p. 5.
- [5] I. Bortins, S. Demers, and K. Clarke, “Cartogram types,” 2002. [Online]. Available: https://web.archive.org/web/20210419121229/http://www.ncgia.ucsb.edu/projects/Cartogram_Central/types.html
- [6] D. Dorling, *Area Cartograms: Their Use and Creation. 1st ed.* University of East Anglia: Environmental Publications, 1996, pp. 1–69.
- [7] D. B. Carr, A. R. Olsen, and D. White, “Hexagon mosaic maps for display of univariate and bivariate geographical data,” *Cartography and Geographic Information Systems*, vol. 19, no. 4, pp. 228–236, 1992. [Online]. Available: <https://doi.org/10.1559/152304092783721231>
- [8] R. G. Cano, K. Buchin, T. Castermans, A. Pieterse, W. Sonke, and B. Speckmann, “Mosaic drawings and cartograms,” *Computer Graphics Forum*, vol. 34, no. 3, p. 361–370, Jun. 2015.
- [9] M. J. Alam, S. G. Kobourov, and S. Veeramoni, “Quantitative measures for cartogram generation techniques,” *Computer Graphics Forum*, vol. 34, no. 3, pp. 351–360, 2015. [Online]. Available: <https://doi.org/10.1111/cgf.12647>
- [10] S. Nickel, M. Sondag, W. Meulemans, M. Chimani, S. Kobourov, J. Peltonen, and M. Nöllenburg, “Computing stable demers cartograms,” in *Graph Drawing and Network Visualization*. Springer-Verlag, 2019, p. 46–60. [Online]. Available: https://doi.org/10.1007/978-3-030-35802-0_4
- [11] R Graph Gallery, “Cartogram,” <https://r-graph-gallery.com/cartogram.html>, accessed: 2026-08-18.
- [12] M. T. Gastner and contributors, “cartogram-cpp,” <https://github.com/mgastner/cartogram-cpp>, 2023.

- [13] S. Nusrat and S. Kobourov, “Visualizing cartograms: Goals and task taxonomy,” 2015. [Online]. Available: <https://doi.org/10.48550/arXiv.1502.07792>
- [14] E. Raisz, “The rectangular statistical cartogram,” *Geographical Review*, vol. 24, no. 2, pp. 292–296, 1934. [Online]. Available: <http://www.jstor.org/stable/208794>
- [15] W. R. Tobler, “A continuous transformation useful for districting,” *Annals of the New York Academy of Sciences*, vol. 219, no. 1, pp. 215–220, 1973. [Online]. Available: <https://doi.org/10.1111/j.1749-6632.1973.tb41401.x>
- [16] S. Nickel, M. Sondag, W. Meulemans, S. Kobourov, J. Peltonen, and M. Nöllenburg, “Multicriteria optimization for dynamic demers cartograms,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 28, no. 6, pp. 2376–2387, 2022. [Online]. Available: <https://doi.org/10.1109/TVCG.2022.3151227>
- [17] W. R. Tobler, “Pseudo-cartograms,” *The American Cartographer*, vol. 13, no. 1, pp. 43–50, 1986. [Online]. Available: <https://doi.org/10.1559/152304086783900194>
- [18] C. J. Kocmoud and D. H. House, “A constraint-based approach to constructing continuous cartograms,” *Digitized Theses and Dissertations (1922–2004)*, 1997. [Online]. Available: <https://hdl.handle.net/1969.1/ETD-TAMU-1997-THESIS-K63>
- [19] D. Keim, S. North, and C. Panse, “Cartodraw: a fast algorithm for generating contiguous cartograms,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 10, no. 1, pp. 95–110, 2004. [Online]. Available: <https://doi.org/10.1109/TVCG.2004.1260761>
- [20] D. Keim, C. Panse, and S. North, “Medial-axis-based cartograms,” *IEEE Computer Graphics and Applications*, vol. 25, no. 3, pp. 60–68, 2005. [Online]. Available: <https://doi.org/10.1109/MCG.2005.64>
- [21] M. T. Gastner and M. E. J. Newman, “Diffusion-based method for producing density-equalizing maps,” *Proceedings of the National Academy of Sciences*, vol. 101, no. 20, pp. 7499–7504, 2004. [Online]. Available: <https://doi.org/10.1073/pnas.0400280101>
- [22] M. T. Gastner, V. Seguy, and P. More, “Fast flow-based algorithm for creating density-equalizing map projections,” *Proceedings of the National Academy of Sciences*, vol. 115, no. 10, pp. E2156–E2164, 2018. [Online]. Available: <https://doi.org/10.1073/pnas.1712674115>
- [23] J.-H. Kämper, S. G. Kobourov, and M. Nöllenburg, “Circular-arc cartograms,” in *2013 IEEE Pacific Visualization Symposium (PacificVis)*, 2013, pp. 1–8. [Online]. Available: <https://doi.org/10.1109/PacificVis.2013.6596121>
- [24] S. Sun, “An optimized rubber-sheet algorithm for continuous area cartograms,” *The Professional Geographer*, vol. 65, no. 1, pp. 16–30, 2013. [Online]. Available: <https://doi.org/10.1080/00330124.2011.639613>
- [25] —, “A fast free-form rubber-sheet algorithm for contiguous area cartograms,” *International Journal of Geographic Information Science*, vol. 27, pp. 567–593, 03 2013. [Online]. Available: <https://doi.org/10.1080/13658816.2012.709247>
- [26] W. Evans, S. Felsner, M. Kaufmann, S. G. Kobourov, D. Mondal, R. I. Nishat, and K. Verbeek, “Table cartograms,” in *Algorithms – ESA 2013*, H. L. Bodlaender and G. F. Italiano, Eds. Springer Berlin Heidelberg, 2013, pp. 421–432.

- [27] J. M. Olson, “Noncontiguous area cartograms,” *The Professional Geographer*, vol. 28, no. 4, pp. 371–380, 1976. [Online]. Available: <https://doi.org/10.1111/j.0033-0124.1976.00371.x>
- [28] R. Heilmann, D. A. Keim, C. Panse, and M. Sips, “Recmap : Rectangular map approximations,” in *InfoVis 2004 : 10th Anniversary : IEEE Symposium on Information Visualization*, M. Ward, Ed. IEEE, 2004, pp. 33–40. [Online]. Available: <https://doi.org/10.1109/INFVIS.2004.57>
- [29] K. Buchin, B. Speckmann, and S. Verdonschot, “Evolution strategies for optimizing rectangular cartograms,” in *Geographic Information Science*. Springer, 2012, pp. 29–42, 7th International Conference on Geographic Information Science. [Online]. Available: https://doi.org/10.1007/978-3-642-33024-7_3
- [30] M. J. Alam, T. Biedl, S. Felsner, M. Kaufmann, S. G. Kobourov, and T. Ueckerdt, “Computing cartograms with optimal complexity,” *Discrete Comput. Geom.*, vol. 50, no. 3, p. 784–810, Oct. 2013. [Online]. Available: <https://doi.org/10.1007/s00454-013-9521-1>
- [31] M. van Kreveld and B. Speckmann, “On rectangular cartograms,” in *Algorithms – ESA 2004*, S. Albers and T. Radzik, Eds. Springer Berlin Heidelberg, 2004, pp. 724–735.
- [32] B. Speckmann, M. {Kreveld, van}, and S. Florisson, “A linear programming approach to rectangular cartograms,” in *Progress in Spatial Data Handling (Proceedings 12th International Symposium)*. Germany: Springer, 2006, pp. 529–546. [Online]. Available: https://doi.org/10.1007/3-540-35589-8_34
- [33] J. A. Dougenik, N. R. Chrisman, and D. R. Niemeyer, “An algorithm to construct continuous area cartograms,” *The Professional Geographer*, vol. 37, no. 1, pp. 75–81, 1985. [Online]. Available: <https://doi.org/10.1111/j.0033-0124.1985.00075.x>
- [34] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004.
- [35] J. O’Rourke, C.-B. Chien, T. Olson, and D. Naddor, “A new linear algorithm for intersecting convex polygons,” *Computer Graphics and Image Processing*, vol. 19, no. 4, pp. 384–391, 1982. [Online]. Available: [https://doi.org/10.1016/0146-664X\(82\)90023-5](https://doi.org/10.1016/0146-664X(82)90023-5)
- [36] G. Toussaint, “A simple linear algorithm for intersecting convex polygons,” *The Visual Computer* 1, 1985. [Online]. Available: <https://doi.org/10.1007/BF01898355>
- [37] M. I. Shamos, “Geometric complexity,” in *Proceedings ACM STOC, 1975*, 1975, pp. 224–233.
- [38] M. I. Shamos and D. Hoey, “Geometric intersection problems,” in *Proceedings IEEE FOCS*, 1976, pp. 208–215.
- [39] W. Meulemans, M. Sondag, and B. Speckmann, “A simple pipeline for coherent grid maps,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 27(2), pp. 1236–1246, 2021. [Online]. Available: <https://doi.org/10.1109/TVCG.2020.3028953>
- [40] D. A. Keim, S. C. North, C. Panse, and J. Schneidewind, “Visualizing geographic information: Visualpoints vs cartodraw,” *Information visualization*, vol. 2, no. 1, pp. 58–67, 2003. [Online]. Available: <https://doi.org/10.1057/palgrave.ivs.9500039>
- [41] A. Meister, “Generalia de genesi figurarum planarum et inde pendentibus earum affectionibus,” *Novi Commentarii Societatis Regiae Scientiarum Gottingensis*, vol. 1, 1769. [Online]. Available: <https://www.digitale-sammlungen.de/en/view/bsb10048766?page=5>

- [42] Y. Lee and W. Lim, “Shoelace formula: Connecting the area of a polygon and the vector cross product,” *Mathematics teacher*, vol. 110, no. 8, pp. 631–636, 2017. [Online]. Available: <https://doi.org/10.5951/mathteacher.110.8.0631>
- [43] J. Hadamard, *Leçons de géométrie élémentaire i Géométrie plane*, ser. Cours Gaston Darboux pour la classe de mathématiques. Librairie Armand Colin, 1906, vol. 2.
- [44] M. De Berg, O. Cheong, M. van Kreveld, and M. Overmars, *Computational geometry: algorithms and applications*. Springer Science & Business Media, 2000.
- [45] GADM, “Gadm license,” <https://gadm.org/license.html>, n.d., accessed: 2026-08-05.
- [46] Pareto Software, LLC., “Simplemaps,” <https://simplemaps.com/>, n.d., accessed August 2026.
- [47] National Archives and Records Administration, Office of the Federal Register, “Distribution of electoral votes,” <https://www.archives.gov/electoral-college/allocation>, 2016, , based on apportionment following the 2010 U.S. Census; accessed August 2026.
- [48] countryeconomy.com, “Taiwan GDP – gross domestic product 2024,” <https://countryeconomy.com/gdp/taiwan>, 2025, nominal GDP figure used for Taiwan; accessed August 2026.
- [49] Ministry of Statistics and Programme Implementation (MoSPI), “State-wise SDP, 2011–12 to 2022–23,” https://mospi.gov.in/sites/default/files/press_releases_statements/State_wise_SDP_01_08_2023_Rev.xls, 2023, accessed: 2026-08-05.
- [50] CBS: centraal bureau voor de statistiek, “Bevolking op 1 januari en gemiddeld; geslacht, leeftijd en regio,” <https://www.cbs.nl/nl-nl/cijfers/detail/03759ned>, 2020, accessed August 2026.
- [51] The World Bank Group, “Population, total,” <https://data.worldbank.org/indicator/SP.POP.TOTL>, 2020, population by Nation; accessed August 2026.
- [52] Statistisches Bundesamt (Destatis), “Tabelle: 12411-0042,” <https://genesis.destatis.de/datenbank/online/table/12411-0042/search/s/MTI0MTETMDA0Mg%3D%3D>, 2024, accessed August 2026.
- [53] S. S. Skiena and S. S. Skiena, *The algorithm design manual*. Springer, 1998, vol. 2.

Chapter 7

Appendix

7.1 Quality Criteria

Table 7.1: Full statistical and geographical accuracy metrics, all datasets. $\bar{\epsilon}/\xi$: mean/max cartographic error 4.1, $\overline{SR}$: mean success rate 4.2, Topological Accuracy $\bar{\delta}/\delta_{max}$ 4.6, mean/max Hamming distance 4.3, $\bar{\theta}$: mean angular orientation error ρ 4.4, orthogonal direction error 4.5, Overlap area fraction 4.7, Gap area fraction 4.4.6, Mean vertex count 4.4.7.

Dataset	System	Cartographic error			Shape (Hamming)			Position		Topology / complexity		
		$\bar{\epsilon}$	ξ	$\overline{SR}$	Topology	$\bar{\delta}$	δ_{max}	$\bar{\theta}$	ρ	Overlap	Gap	Complexity
Ideal	Values	0	0	1	0	0	0	0	0	0	0	low
<i>Contiguous</i>												
USA	Gastner cpp	0.016	0.242	1.038	0.045	0.326	0.615	0.398	0.259	0	2.084	129
	R (conti)	0.052	0.964	0.966	0	0.273	0.456	0.187	0.117	0.000	1.858	201
	Framework	0.446	0.989	1.238	0.244	0.248	0.527	0.099	0.077	0	1.212	400
India	Gastner cpp	0.002	0.007	1.015	0	0.296	0.775	0.067	0.047	0.000	2.749	302
	R (conti)	0.107	0.759	1.214	0	0.296	0.616	0.05	0.04	0.000	2.697	151
	Framework	0.339	0.988	0.999	0.262	0.198	0.490	0.099	0.081	0.000	1.839	565
Taiwan & China	Gastner cpp	0.034	0.442	0.973	0	0.381	0.907	0.197	0.109	0	3.987	271
	R (conti)	0.204	0.904	1.139	0.879	0.370	0.904	0.19	0.121	0.000	2.970	95
	Framework	0.522	0.966	1.447	0.398	0.24	0.692	0.209	0.130	0	1.063	889
Netherlands	Gastner cpp	0.001	0.004	0.998	0	0.272	0.406	0.207	0.030	0.000	1.838	551
	R (conti)	0.031	0.126	0.94	0	0.189	0.292	0.08	0.015	0.000	1.831	41
	Framework	0.133	0.393	1.028	0.217	0.164	0.344	0.053	0.015	0	1.679	1267
Germany	Gastner cpp	0.076	0.572	1.031	0	0.243	0.380	0.184	0.42	0.042	2.144	605
	R (conti)	0.124	0.581	0.935	0	0.199	0.364	0.083	0.042	0.034	2.107	43
	Framework	0.161	0.781	1.06	0.1	0.145	0.376	0.086	0.033	0	1.321	1094
World	Gastner cpp	0.134	1	1.086	0	0.491	1	0.33	0.141	0.007	5.777	43
	R (conti)	0.333	1	1.269	0.007	0.394	1	0.143	0.096	0.000	3.659	102
	Framework	0.556	0.999	1.488	0.694	0.25	0.947	0.174	0.113	0.039	1.135	71
<i>Dorling</i>												
USA	R (dorling)	0.000	0.000	1	1	0.257	0.467	0.395	0.210	0	2.282	120
	Framework	0.000	0.000	1	1.0	0.257	0.467	0.205	0.202	0	2.094	64
India	R (dorling)	0.071	0.147	1.261	1	0.263	0.492	0.044	0.025	0	1.502	120
	Framework	0.000	0.000	1	1	0.267	0.491	0.098	0.059	0	1.142	64
Taiwan & China	R (dorling)	0.088	0.213	1.199	1	0.254	0.666	0.154	0.0852	0	1.856	120
	Framework	0.000	0.000	1	1	0.261	0.672	0.238	0.15	0	1.836	64
Netherlands	R (dorling)	0.012	0.025	0.987	1	0.230	0.492	0.004	0	0	2.789	120
	Framework	0.000	0.000	1	1	0.285	0.42	0.236	0.151	0	1.043	64
Germany	R (dorling)	0.033	0.07	0.992	1	0.211	0.316	0.025	0.017	0	3.677	120
	Framework	0.000	0.000	1	1	0.247	0.396	0.179	0.117	0	0.799	64
World	R (dorling)	0.148	0.711	1.241	1	0.313	0.835	0.093	0.062	0	1.908	120
	Framework	0.012	1	1	1	0.301	0.843	0.174	0.108	0.000	2.439	64

Table 7.2: Full statistical and geographical accuracy metrics, all datasets (continued).

Dataset	System	Cartographic error			Shape (Hamming)		Position		Topology / complexity			
		$\bar{\epsilon}$	ξ	$\overline{SR}$	Topology	$\bar{\delta}$	$\delta_{\max}$	$\bar{\theta}$	ρ	Overlap	Gap	Complexity
<i>Non-contiguous</i>												
USA	R (non)	0.012	0.113	0.993	1	0.181	0.35	0.161	0.091	0.097	2.953	199
	Framework	0.001	0.012	1	1	0.000	0.000	0.075	0.059	0.000	0.2	1740
India	R (non)	0.071	0.143	1.259	1	0.030	0.382	0.001	0	0.068	3.274	151
	Framework	0.000	0.001	1	1	0.000	0.000	0.033	0.021	0	0.393	2262
Taiwan & China	R (non)	0.075	0.213	1.162	1	0.039	0.322	0.007	0.006	0.087	5.578	103
	Framework	0.014	0.337	1	1	0.000	0.000	0.063	0.032	0	0.032	1379
Netherlands	R (non)	0.012	0.025	0.987	1	0.033	0.083	0.004	0	0.002	3.353	41
	Framework	0.000	0.000	0.917	1	0.000	0.000	0.050	0.015	0.003	1.304	284
Germany	R (non)	0.035	0.068	0.992	1	0.045	0.242	0.012	0.008	0.031	3.026	43
	Framework	0.000	0.000	1	1	0.000	0.000	0.078	0.05	0	0.380	842
World	R (non)	0.129	0.728	1.338	1	0.037	0.348	0.004	0.004	0.01	8.483	108
	Framework	0.106	1	1	1	0.000	0.000	0.013	0.010	0	0.012	1229
<i>Demers</i>												
USA	Framework	0.000	0.000	1	0.768	0.269	0.505	0.157	0.146	0.000	1.583	4
India	Framework	0.000	0.000	1	0.721	0.283	0.502	0.165	0.114	0.000	1.417	4
Taiwan & China	Framework	0.000	0.000	1	0.797	0.274	0.646	0.247	0.155	0	2.922	4
Netherlands	Framework	0.000	0.000	1	0.739	0.309	0.452	0.287	0.136	0.000	1.232	4
Germany	Framework	0.000	0.000	1	0.759	0.263	0.421	0.205	0.133	0.000	1.030	4
World	Framework	0.000	0.000	1	0.763	0.310	0.801	0.155	0.110	0.000	2.332	4